

CHAPTER 7
HALOALKANES
(ALKYL HALIDES)

7.1 INTRODUCTION TO HALOALKANES

INTRODUCTION

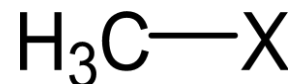
- Haloalkanes are compounds in which a halogen atom (F, Cl, Br or I) replaces a H atom of an alkane.
- General formula:
 $C_nH_{2n+1}X$ or RX

where $X = Cl, Br, F, I$
- The common name of **haloalkanes** is **alkyl halides**.

CLASSIFICATION OF HALOALKANES

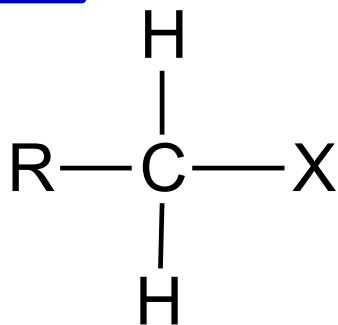
3

METHYL HALIDE



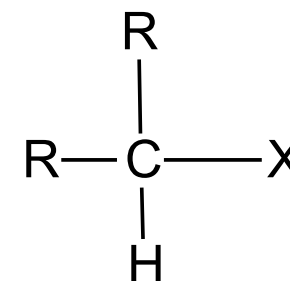
Halogen is bonded to methyl group

PRIMARY HALOALKANE



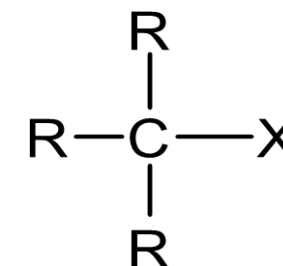
1 alkyl group bonded to α -carbon

SECONDARY HALOALKANE



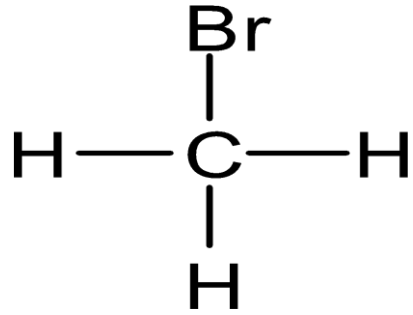
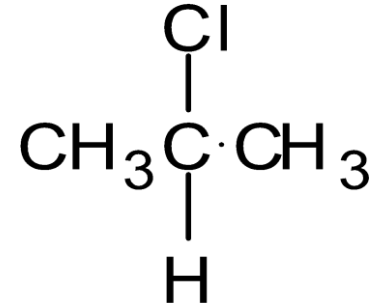
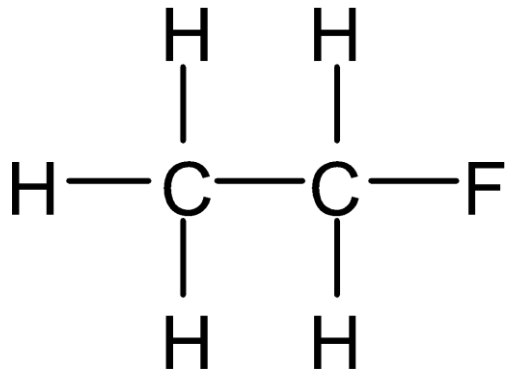
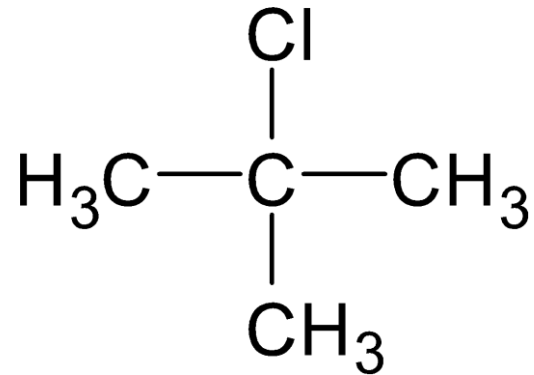
2 alkyl groups bonded to α -carbon

TERTIARY HALOALKANE



3 alkyl groups bonded to α -carbon

Depending on the **number of alkyl groups** attached to the α -carbon. α -carbon refers to the carbon atom bonded to halogen (X)

EXAMPLE :**Methyl halide****2° haloalkane****1° haloalkane****3° haloalkane**

NOMENCLATURE

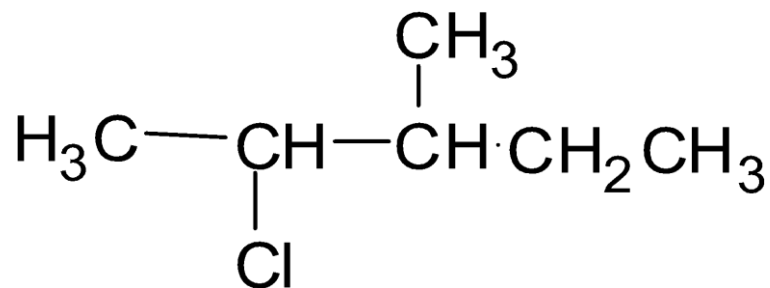
5

- A haloalkane is named as an **alkane with halogen as substituent**.
- The name is based on the **longest carbon chain**.
- To name the halogen substituent, use prefixes *fluoro-*, *chloro-*, *bromo-* or *iodo-*.

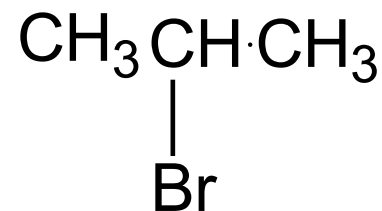
Example:



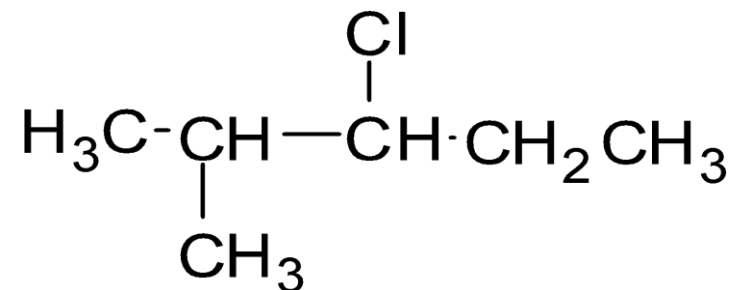
chloroethane



2-chloro-3-methylpentane



2-bromopropane

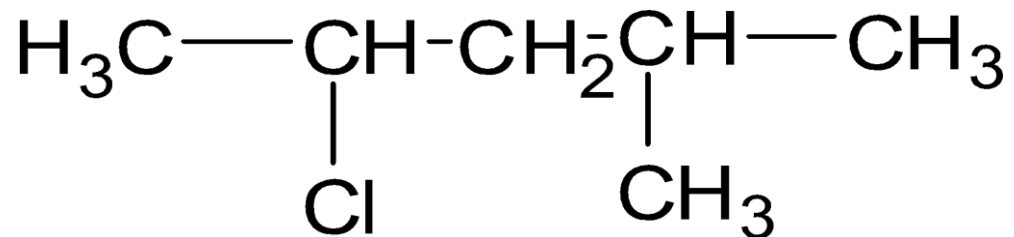


3-chloro-2-methylpentane

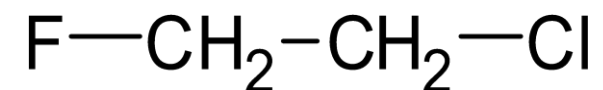
NOMENCLATURE

- If 2 or more substituents (halogen or alkyl groups) are present, find the parent carbon chain.
- Apply other rules for nomenclature
 - i. Number the chain starting at the end nearest the first substituent either alkyl or halogen.
 - ii. Name and number the substituent.
 - ii. Alphabetise the substituents.

Example:



2-chloro-4-methylpentane

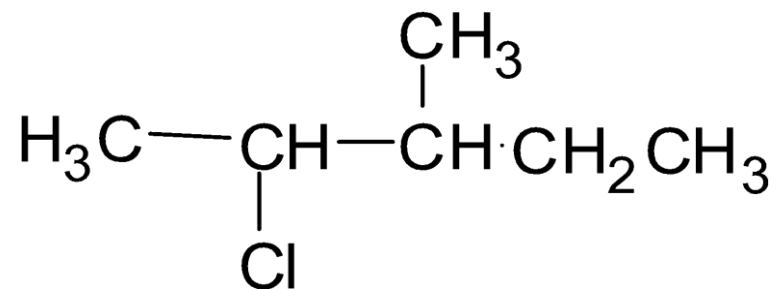


1-chloro-2-fluoroethane

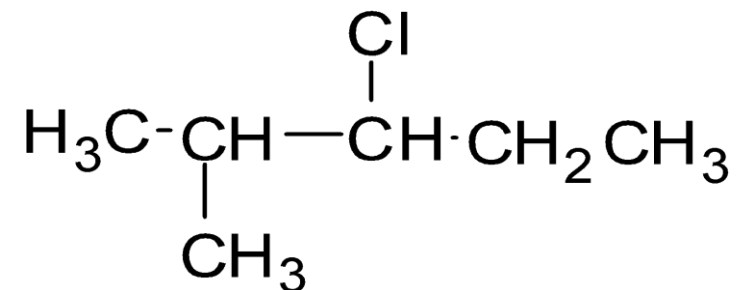
NOMENCLATURE

7

Example:



2-chloro-3-methylpentane

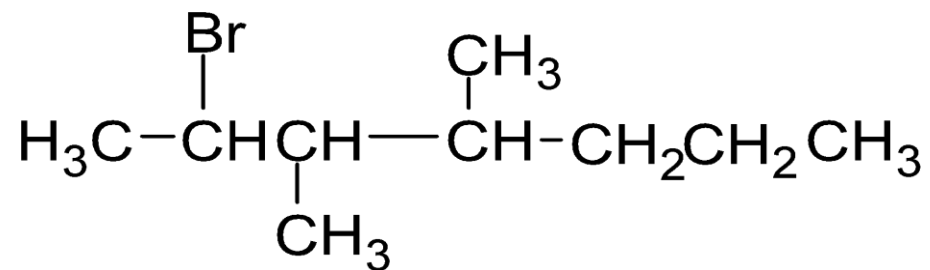


3-chloro-2-methylpentane

Exercise:

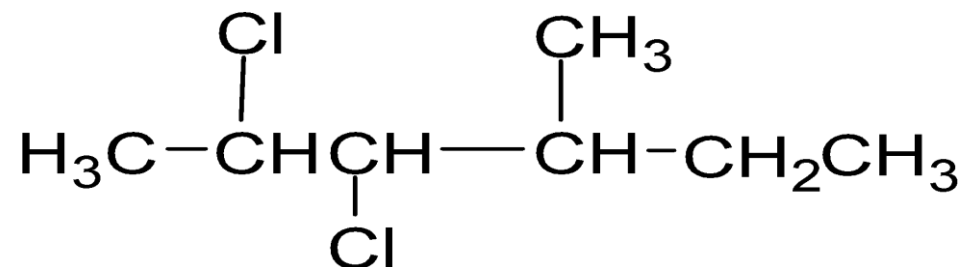
Give IUPAC names for the following compounds:

①



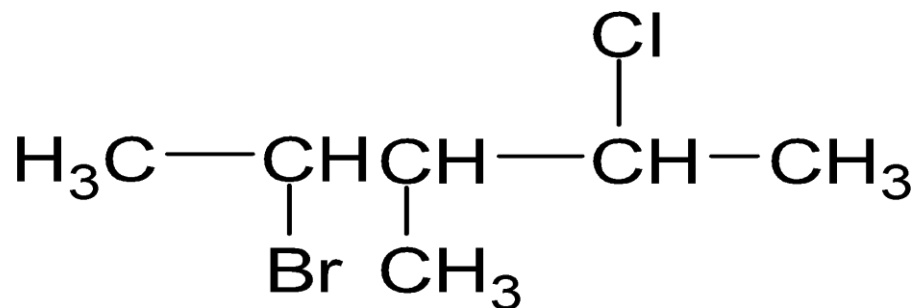
2-bromo-3,4-dimethylheptane

②

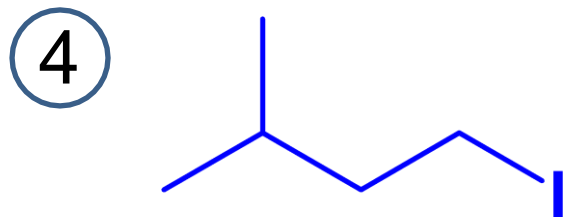


2,3-dichloro-4-methylhexane

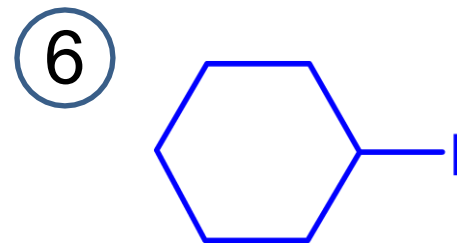
③



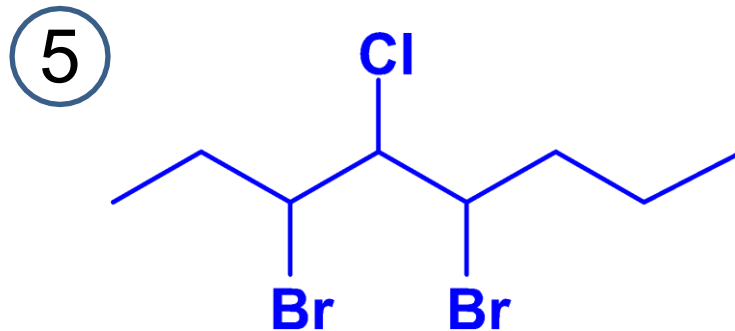
2-bromo-4-chloro-3-methylpentane



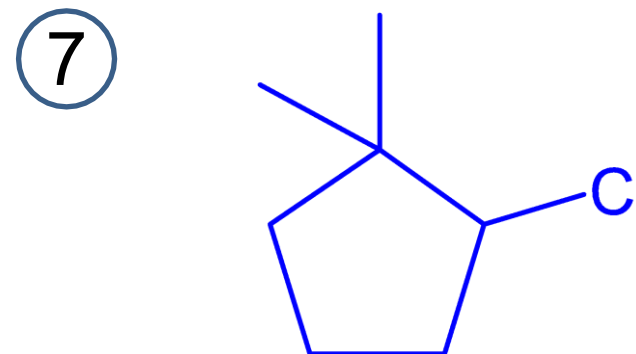
1-iodo-3-methylbutane



iodocyclohexane



3,5-dibromo-4-chlorooctane

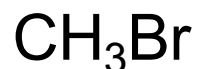


2-chloro-1,1-dimethylcyclopentane

COMMON NAME

- Common names of haloalkanes are constructed by naming the alkyl group and then the halide $\Rightarrow$ **alkyl halide**.
- Common names are useful only for simple alkyl halides.

Example:

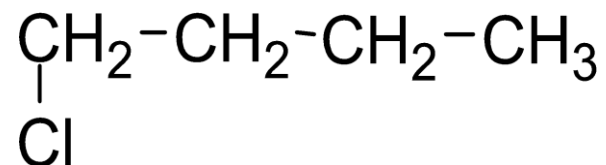


**Common
name**

methyl bromide

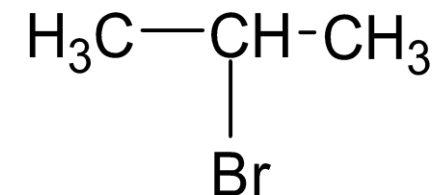
**IUPAC
name**

bromomethane



n-butyl chloride

1-chlorobutane

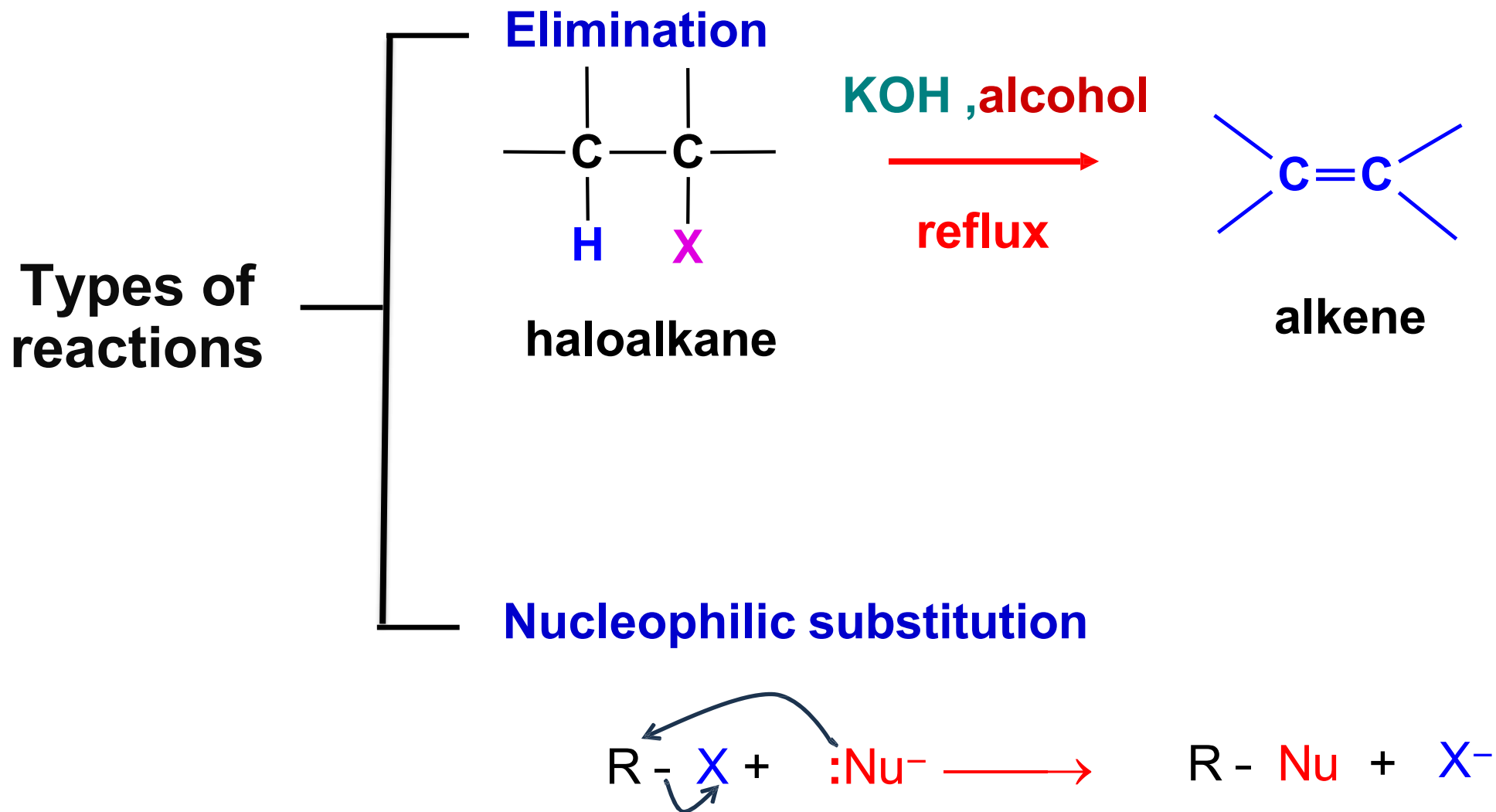


isopropyl bromide

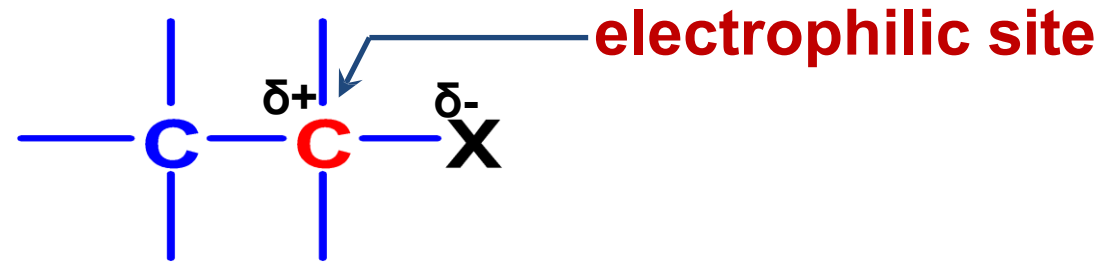
2-bromopropane

7.2 CHEMICAL PROPERTIES OF HALOALKANES

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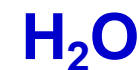
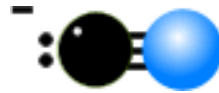


- Halogens have electronegativities significantly greater than C atom.
- Consequently, the C - X bond is polarised.

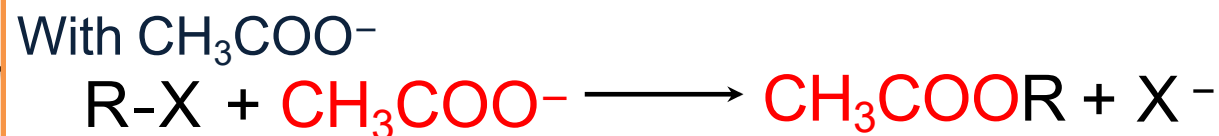
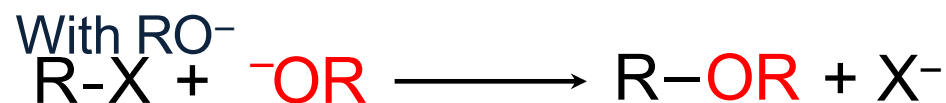
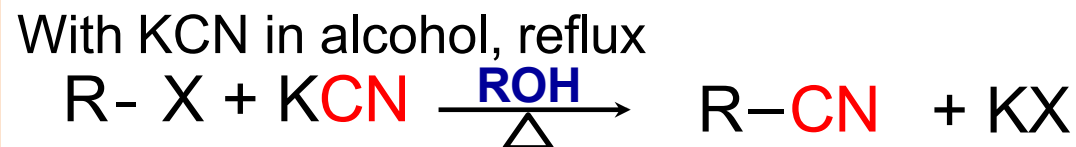
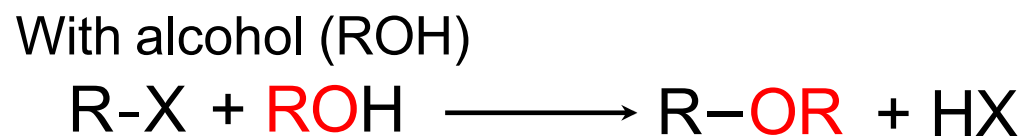
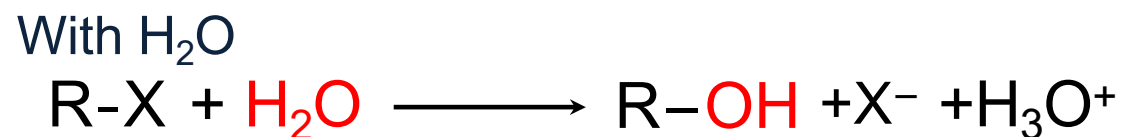
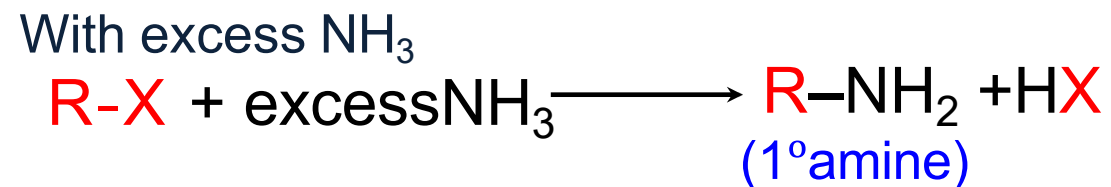


- The electrophilic C atom is susceptible to nucleophilic attack.
- Thus, haloalkanes tend to undergo **nucleophilic substitution reaction** involving the **replacement of the halogen by a nucleophile**.
- The nucleophile is either an **anion** or a **neutral molecule** with lone pair at the central atom.

e.g.



Nucleophilic Substitution Reactions of R-X



1. REACTION OF HALOALKANE WITH AQUEOUS NaOH

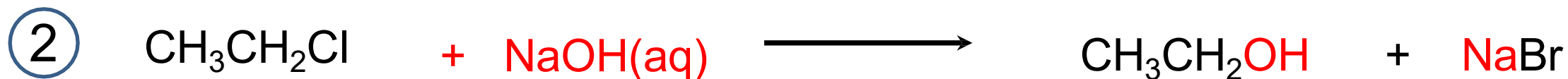
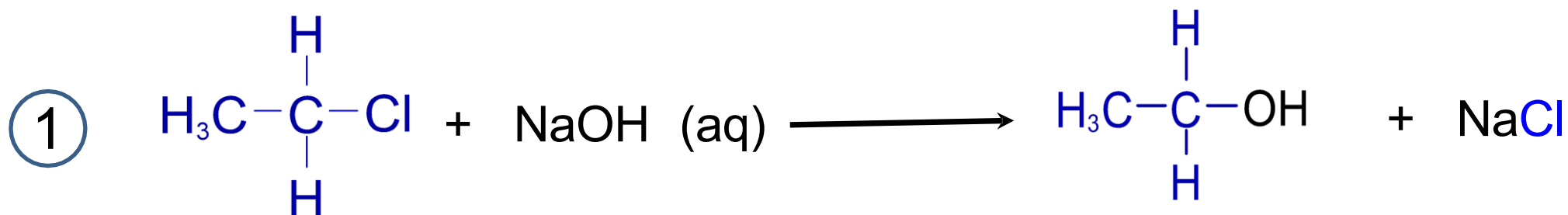
14

- When haloalkane is treated with aqueous NaOH or KOH, halogen is replaced by -OH to produce alcohol.

General reaction:



EXAMPLE :

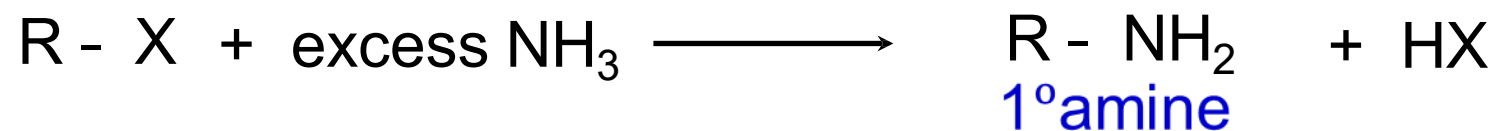


2. REACTION OF HALOALKANE WITH EXCESS NH_3

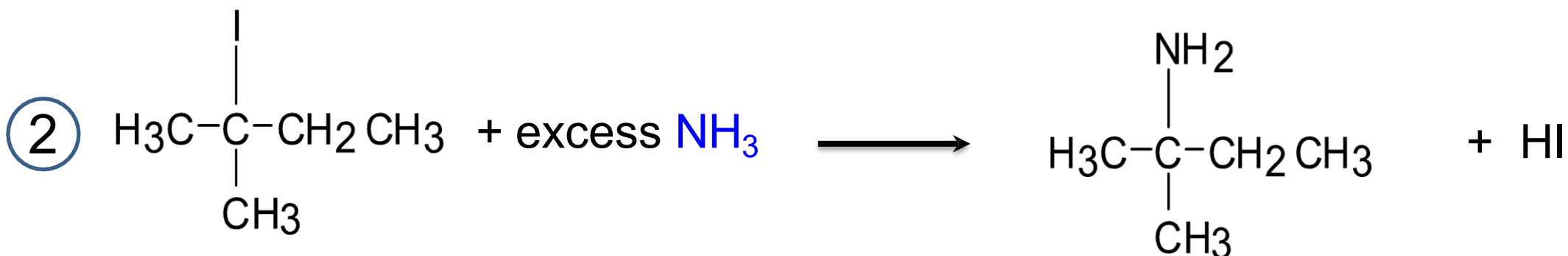
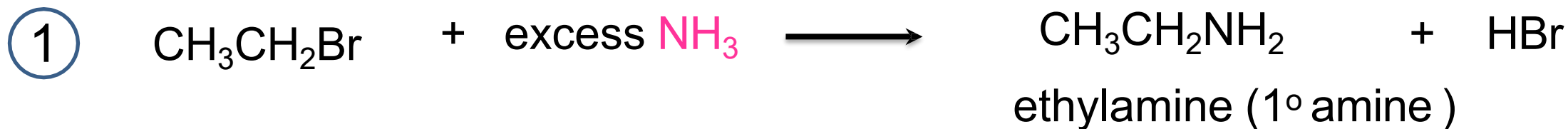
15

- Haloalkane is heated under reflux with excess ammonia.
- Halogen is replaced by $-\text{NH}_2$ to produce 1° amine.

General reaction:



Example:



3. REACTION OF HALOALKANE WITH WATER (HYDROLYSIS)

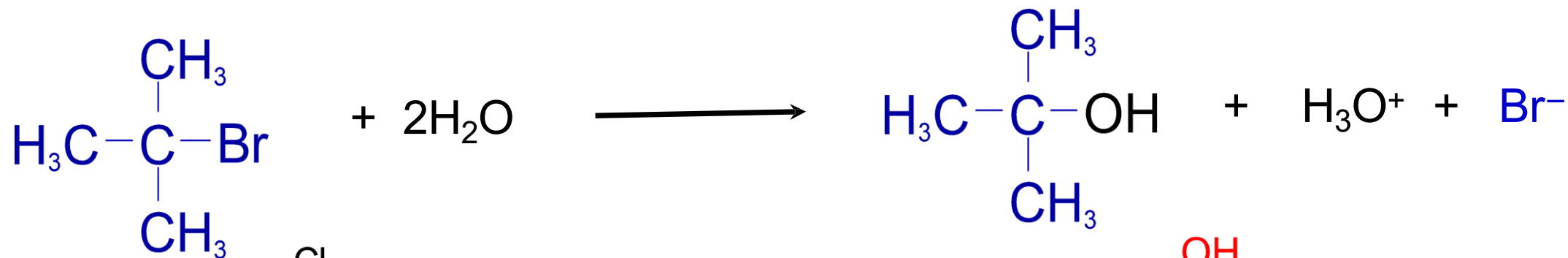
- Haloalkane reacts with water, H_2O to produce alcohol.
- Favoured by 3° haloalkane via $\text{S}_\text{N}1$ mechanism.

General equation:

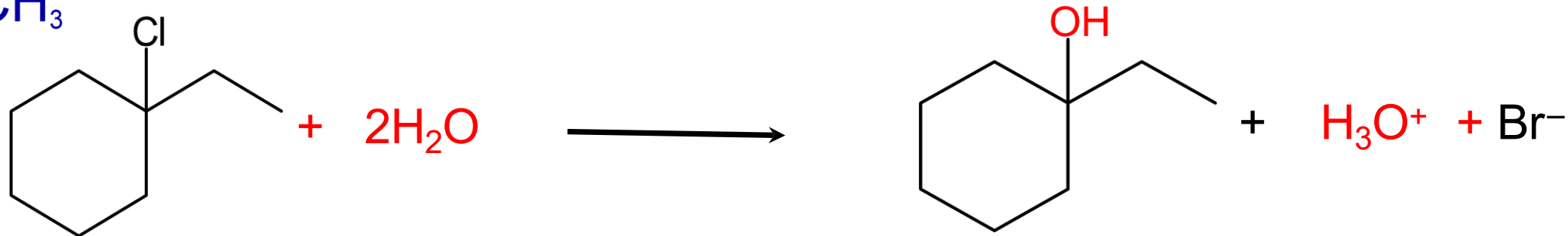


EXAMPLE :

①



②

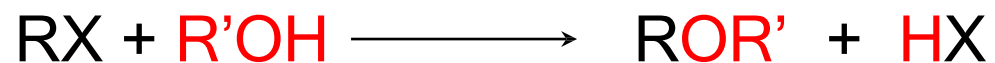


4. REACTION OF HALOALKANE WITH ALCOHOL (ALCOHOLYSIS)

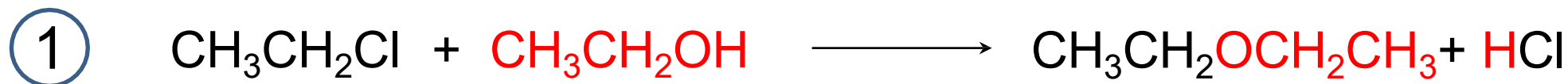
17

- Haloalkane is reacted with alcohol
- Halogen is replaced by -OR to produce ether

General reaction:



EXAMPLE :



5. REACTION OF HALOALKANE WITH ALCOHOLIC KCN @ NaCN

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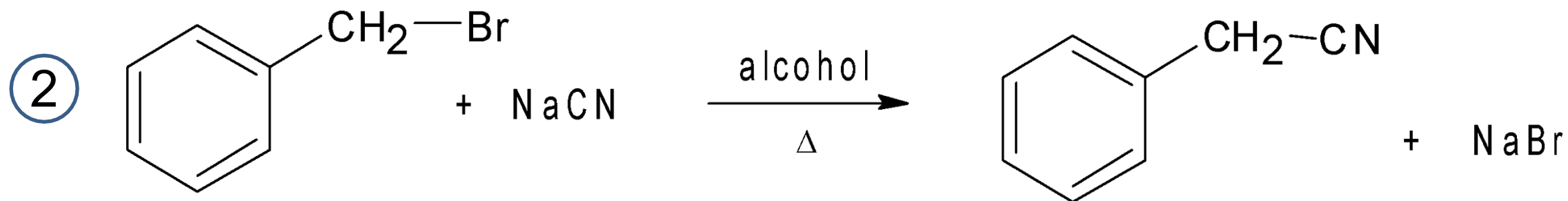
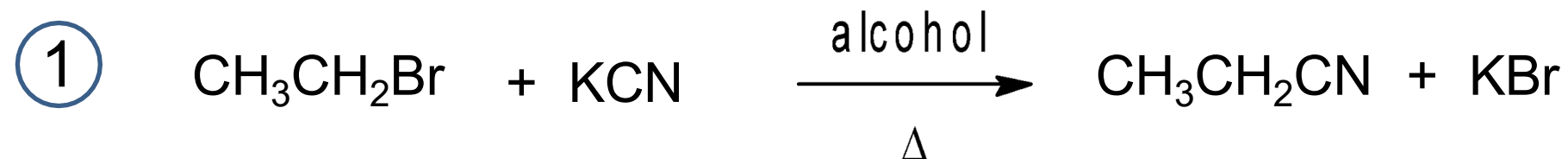
- Haloalkane is heated under reflux with solution of potassium or sodium cyanide in alcohol.
- Halogen is replaced by -CN to produce nitrile compound.

General reaction:



- In this reaction, number of C atom in the product is increased by one (lengthening the chain)

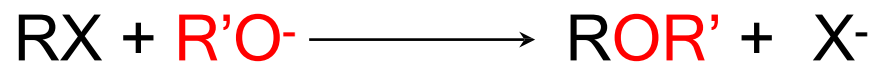
EXAMPLE :



6. REACTION OF HALOALKANE WITH SODIUM OR POTASSIUM ALKOXIDE

- When haloalkane reacts with potassium or sodium alkoxide, halogen is replaced by $-OR$ to produce ether.
- Normally occur to 1° haloalkane.

General reaction:



EXAMPLE :

- $$CH_3CH_2CH_2Br + CH_3CH_2ONa \longrightarrow CH_3CH_2CH_2OCH_2CH_3 + NaBr$$
- $$CH_3CH_2Cl + CH_3O^- \longrightarrow CH_3CH_2OCH_3 + Cl^-$$

7. REACTION OF HALOALKANE WITH SODIUM OR POTASSIUM CARBOXYLATE

- When haloalkane reacts with sodium or potassium carboxylate, halogen is replaced by RCOO^- to produce ester

General reaction:



EXAMPLE :



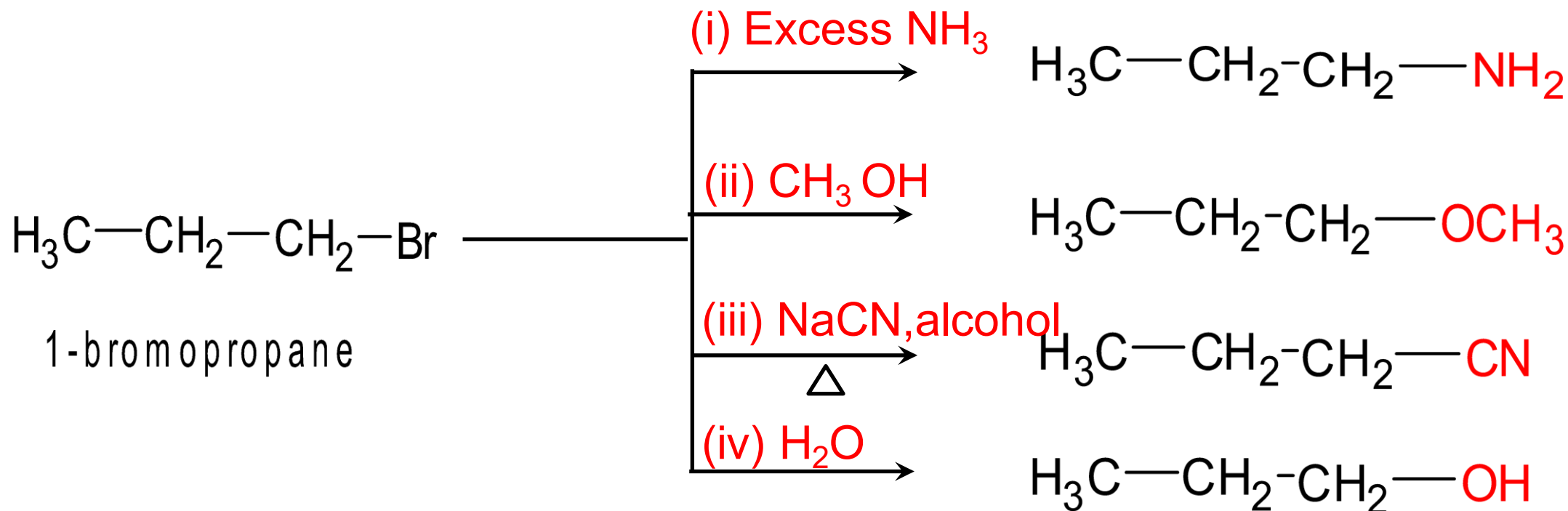
EXERCISE

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Draw the structure of the product formed when 1-bromopropane reacts with the following reagents respectively:

- (i) Excess Ammonia
- (ii) Methanol
- (iii) Sodium cyanide
- (iv) Water

ANSWER



MECHANISM OF REACTION OF R-X

NUCLEOPHILIC SUBSTITUTION

2 types of mechanism for nucleophilic substitution reaction:

- Unimolecular Nucleophilic Substitution Reaction, S_N1
- Bimolecular Nucleophilic Substitution Reaction, S_N2

The mechanism involved in nucleophilic substitution depends on:

- the structure of the haloalkane
- strength of nucleophile

UNIMOLECULAR NUCLEOPHILIC SUBSTITUTION, S_N1

- First order reaction
- Rate equation :
$$\text{rate} = k [\text{haloalkane}]$$
- Rate of reaction depends only on the **concentration of haloalkane**.
- Favoured by 3° RX and 1° RX with large (bulky) alkyl groups.
- Usually involves **weak nucleophiles**.
- S_N1 is **multi-step mechanism** and involves the formation of carbocation.

STEP 1: Formation of the carbocation

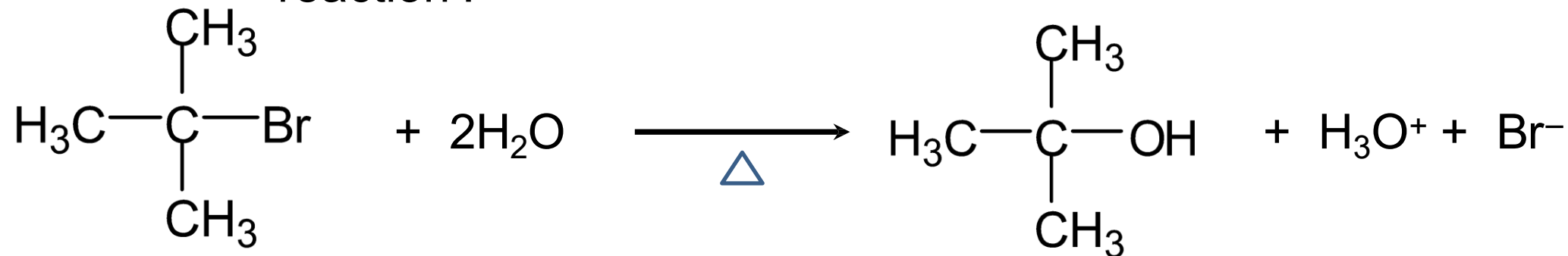
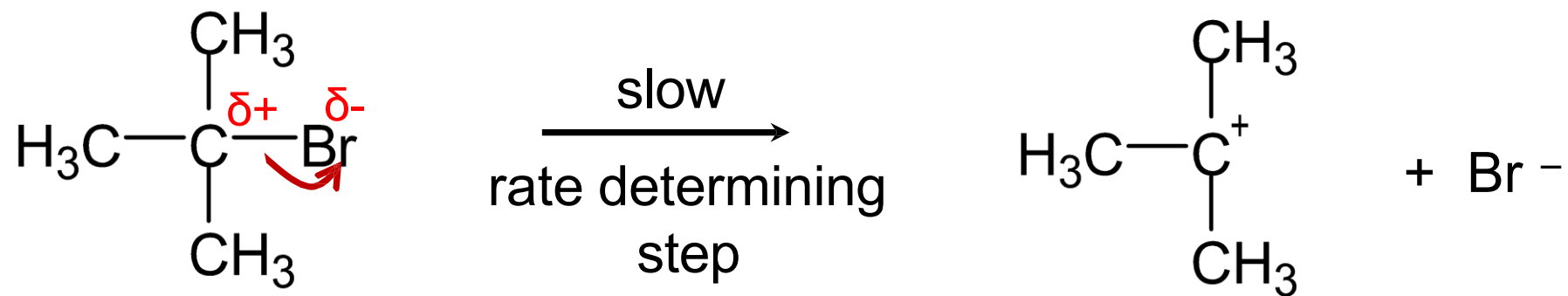
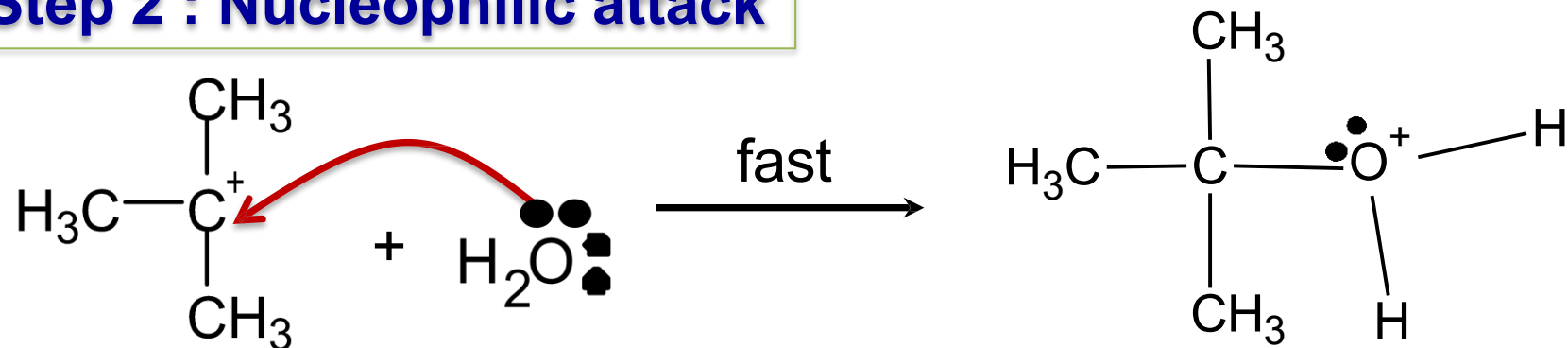
STEP 2: Nucleophilic attack

STEP 3: Loss of proton or deprotonation

EXAMPLE 1:

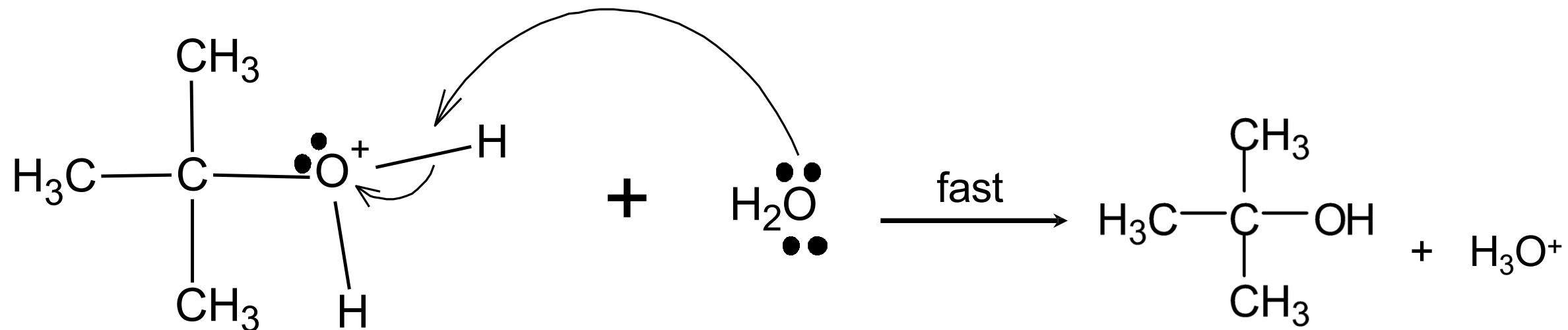
Write a **complete mechanism** for the following **hydrolysis** reaction :

25

**Step 1 : Formation of carbocation****Step 2 : Nucleophilic attack**

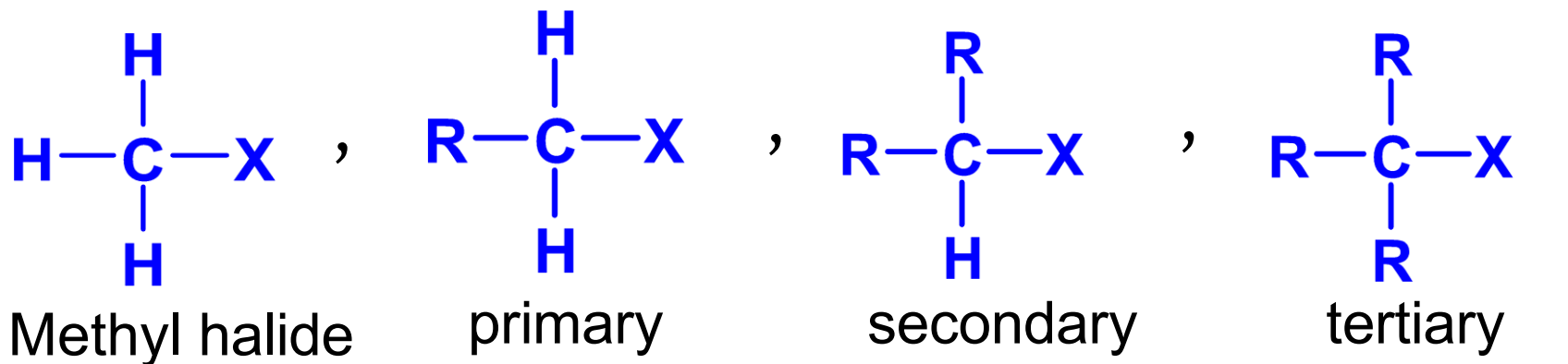
Step 3 : Deprotonation

26

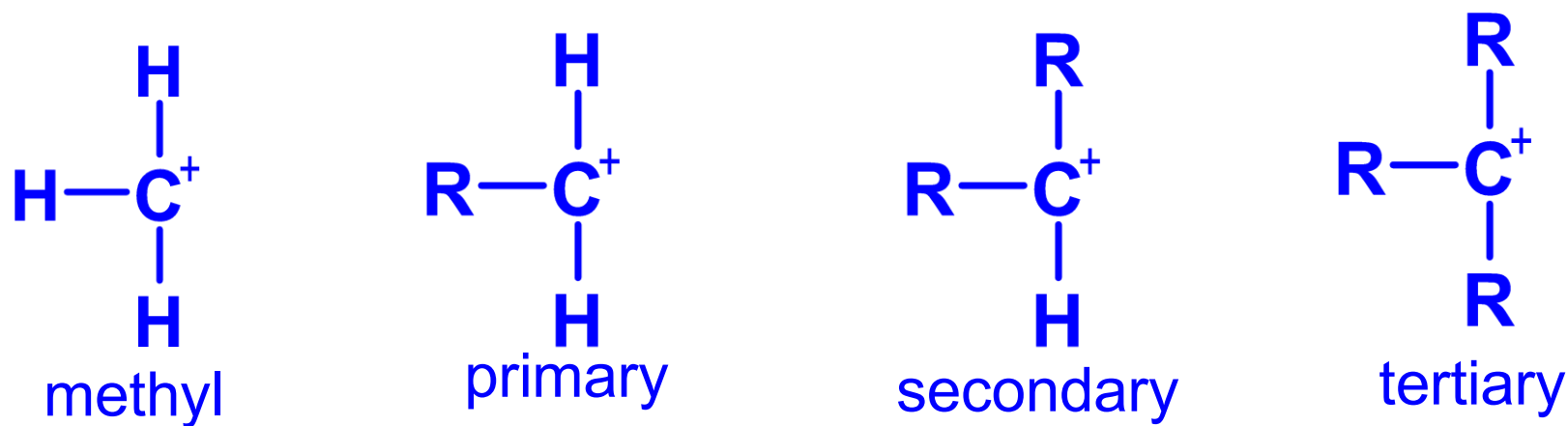


RELATIVE REACTIVITIES OF 1°, 2° AND 3° R-X OF S_N1

- For RX with the same halogen, the **relative reactivity of haloalkanes towards S_N1** reaction corresponds to the **relative carbocation stability**.

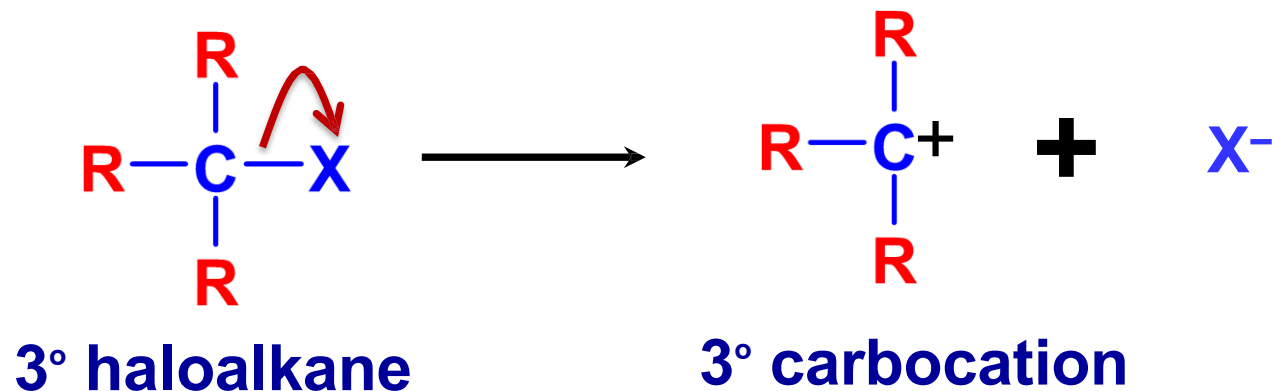


Increasing reactivity of haloalkanes towards S_N1 reaction



Increasing stability of carbocation

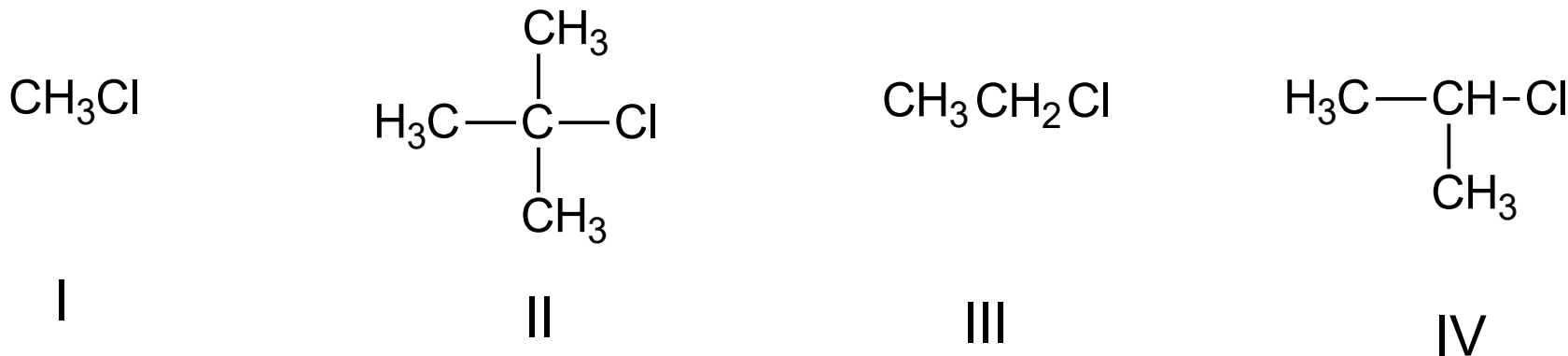
- Factor determines reactivity of RX of S_N1 reaction is the **stability of the carbocation**.
- 3° RX undergo S_N1 reactions the most rapidly because **3° carbocation is the most stable**.



- Alkyl group is **electron-donating group**.
- More alkyl groups attached to the positively charged carbon will stabilise the carbocations through **inductive effect**.
- Thus, **3° carbocation is the most stable compared to 2° and 1° carbocations**.

EXAMPLE:

Arrange the following haloalkane in order of decreasing reactivity towards S_N1 hydrolysis. Explain your answer.

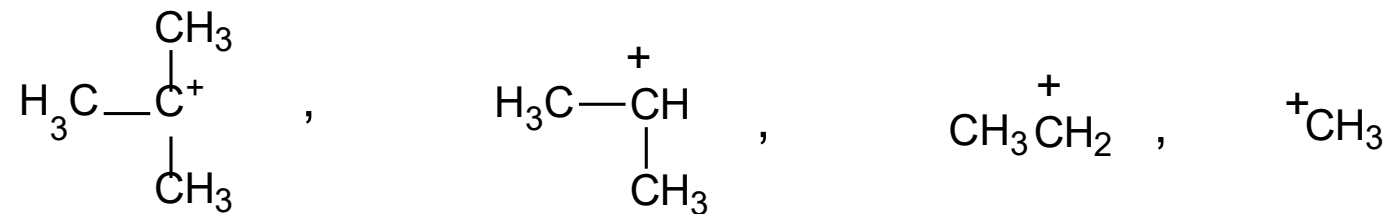


ANSWER:

II, IV, III, I

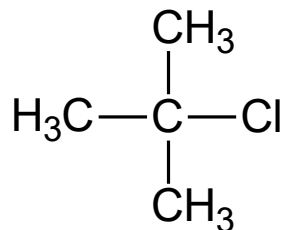
—————→
decreasing reactivity towards S_N1 hydrolysis

The more stable the carbocation intermediate formed, the more susceptible it is to nucleophilic attack, hence the more reactive the haloalkane towards S_N1 reaction.



decreasing stability of carbocation

Structure II is a **tertiary haloalkane** which forms the **most stable tertiary carbocation**. Therefore, I is the most reactive.



II

Structure I is **methyl halide** that forms the **least stable methyl cation @ methylium ion**. Therefore, it is the least reactive.

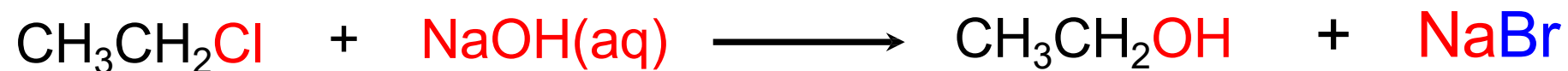


BIMOLECULAR NUCLEOPHILIC SUBSTITUTION, S_N2

- **Second order reaction**
- Rate equation :
Rate = k [haloalkane] [nucleophile]
- Rate of reaction depends on the **concentration of haloalkane** and **concentration of nucleophile**
- **Favoured by methyl halide and 1° RX.**
- The mechanism has only **ONE STEP.**
- Involved formation **of transition state.**
- The processes of **bond breaking** and **bond forming** occur **simultaneously** as one bond is forming, while the other bond is breaking.

EXAMPLE:

Consider the following **alkaline hydrolysis reaction** :

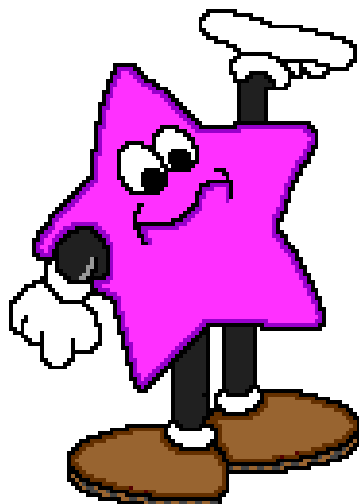


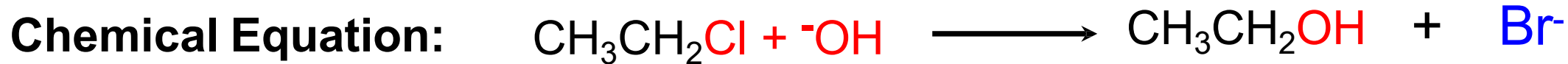
1°RX



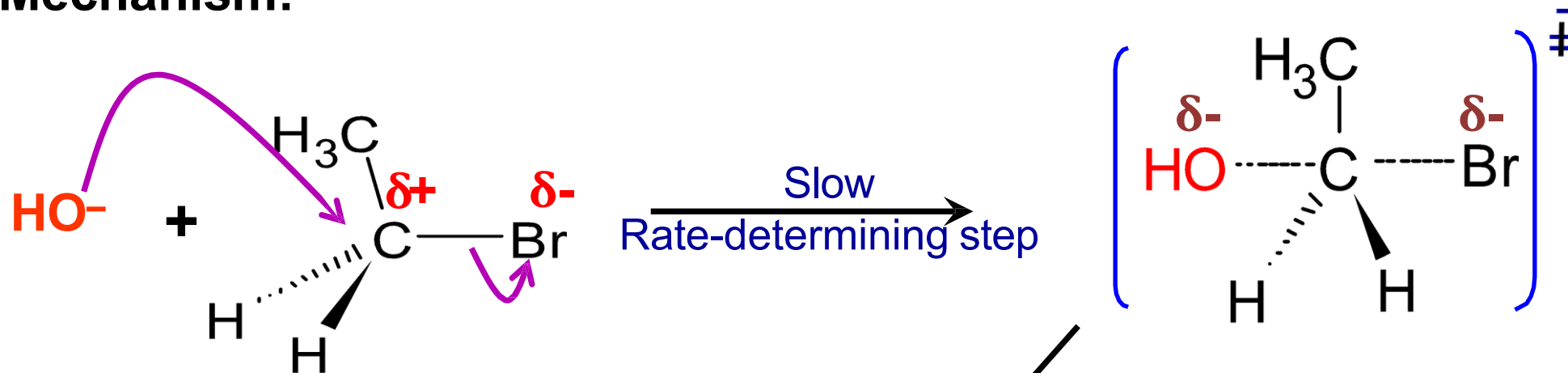
OH^-

Nucleophile



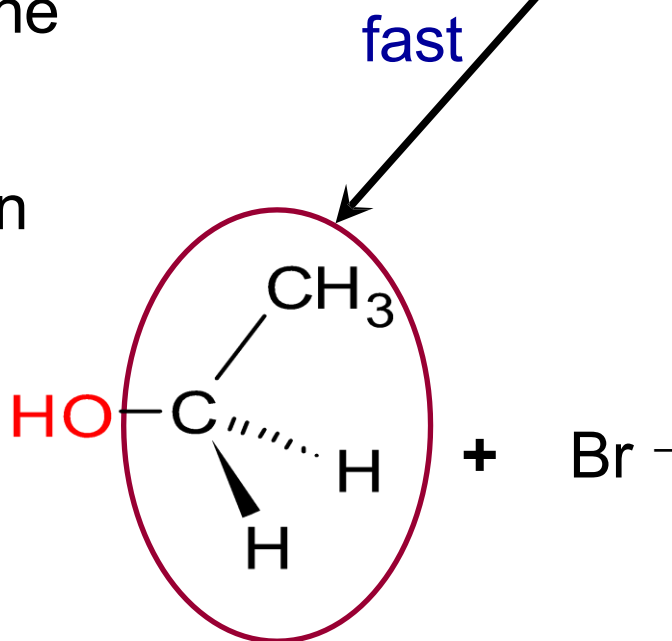


$\text{S}_{\text{N}}2$ Mechanism:



(i) OH^- ion (nucleophile) attacks the electrophilic α -carbon atom

(ii) To minimise repulsions between RX and OH^- , the OH^- attacks the electrophilic C atom at the side opposite to the halogen.



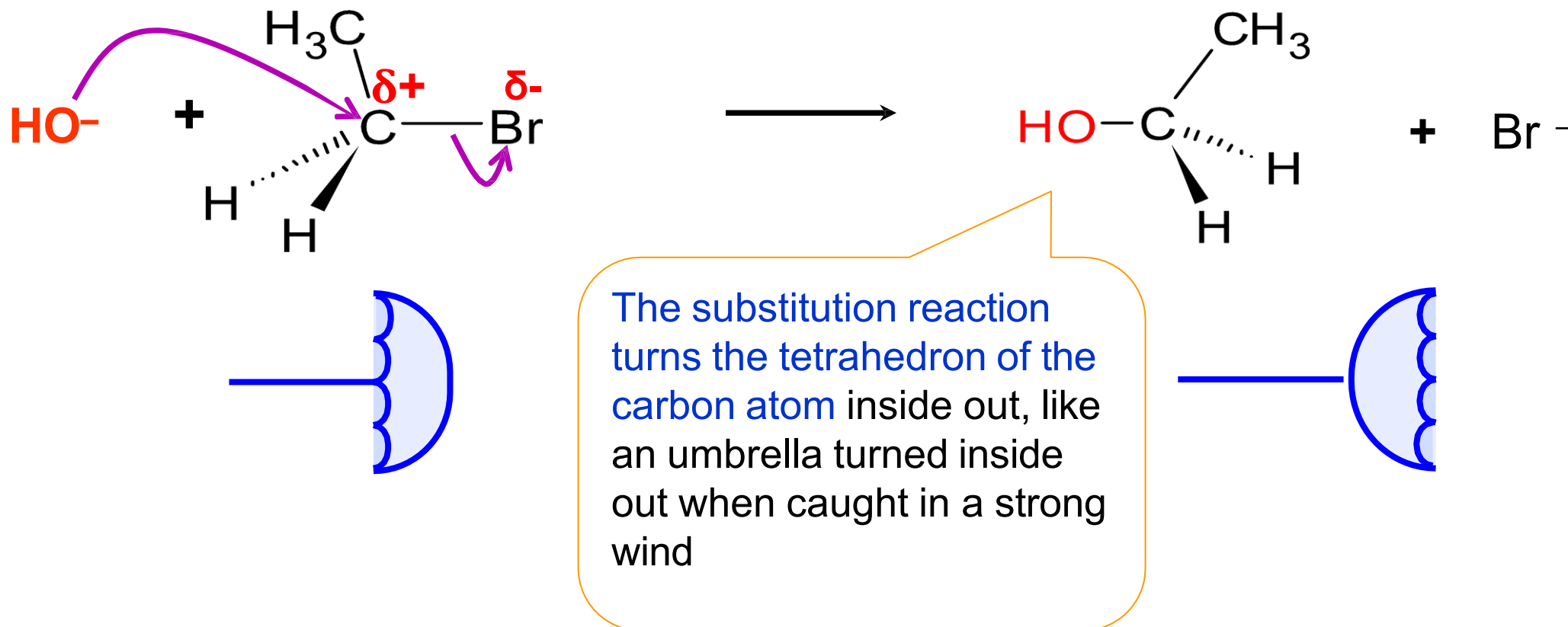
Transition state

(iii) When the $\text{C}-\text{OH}$ bond begins to form, the $\text{C}-\text{Br}$ bond becomes weakened and finally broken.

(iv) Br is replaced by OH and the configuration of product is inverted.

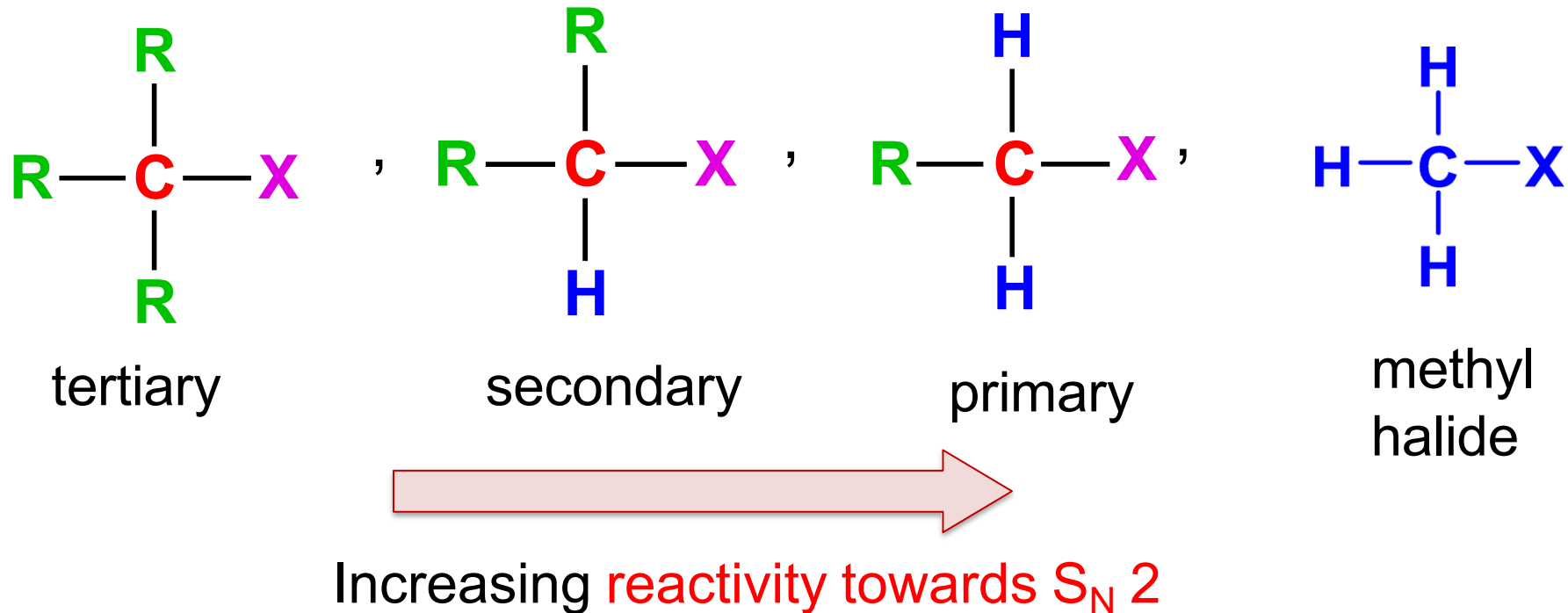
STEREOCHEMISTRY IN S_N2

- All S_N2 reactions proceed with **backside attack by the nucleophile**, hence the **product formed has inverse configuration**.



Relative Reactivities of 1°, 2° and 3° RX with S_N2

- For RX with the **same halogen**



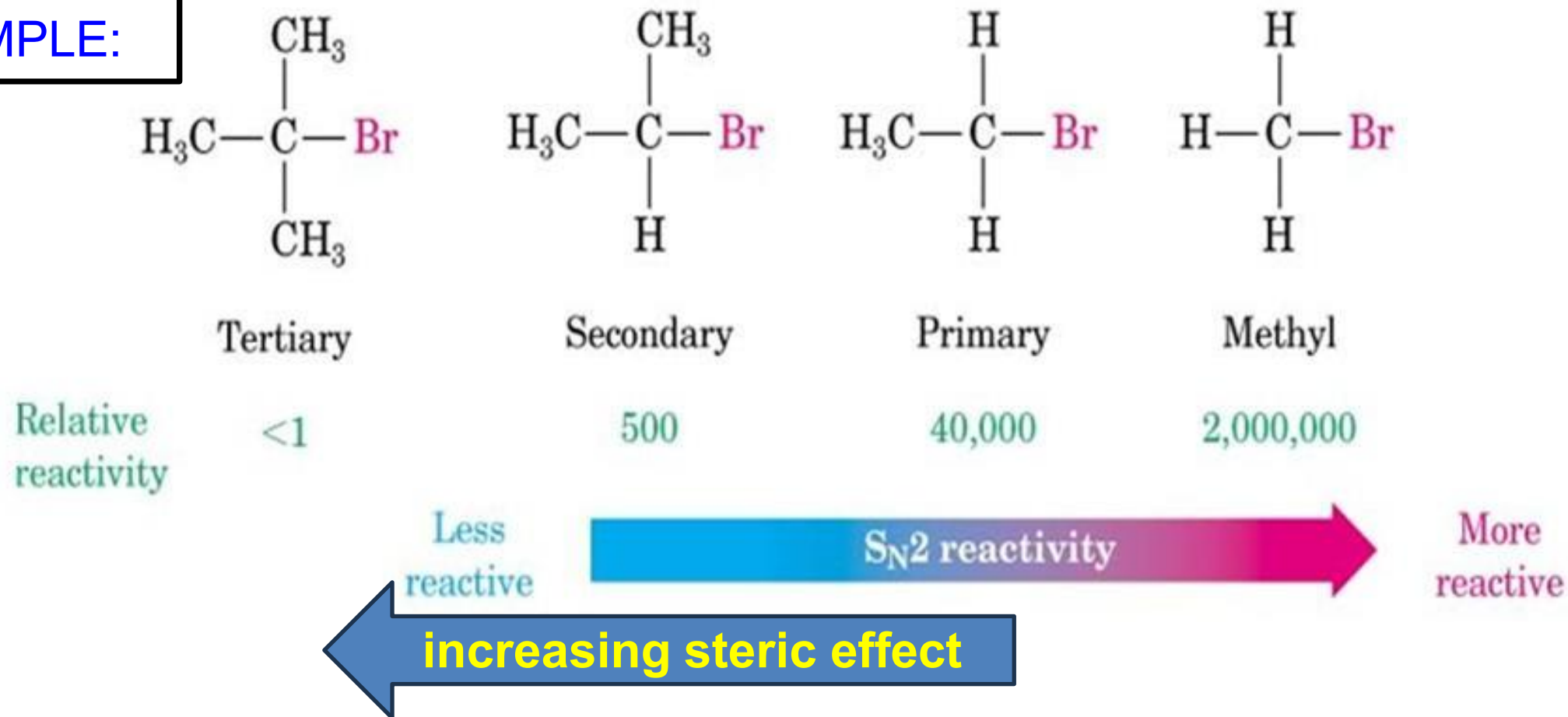
- Factor that determines the order of reactivity in S_N2 is **steric effect**.
- If **steric effect increases**, **reactivity** of haloalkane will **decrease**.

STERIC EFFECT IN S_N2



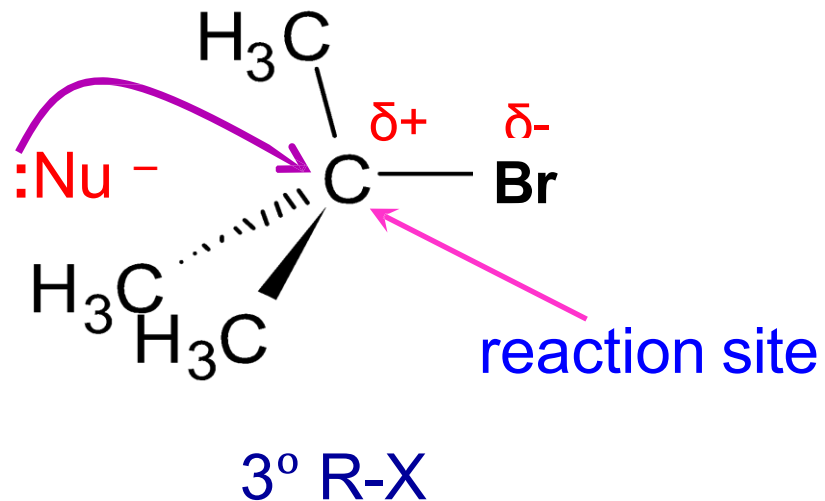
- Steric effect is an effect in which the rate of reaction depends on the **spatial arrangement** of the group attached to, or near to the reaction site of the molecule.
- Steric effect caused by bulky alkyl groups at reaction site (α -carbon) makes the attack of nucleophile (or **nucleophilic attack**) more difficult.
- **Steric effect** is also known as **steric hindrance**.

EXAMPLE:

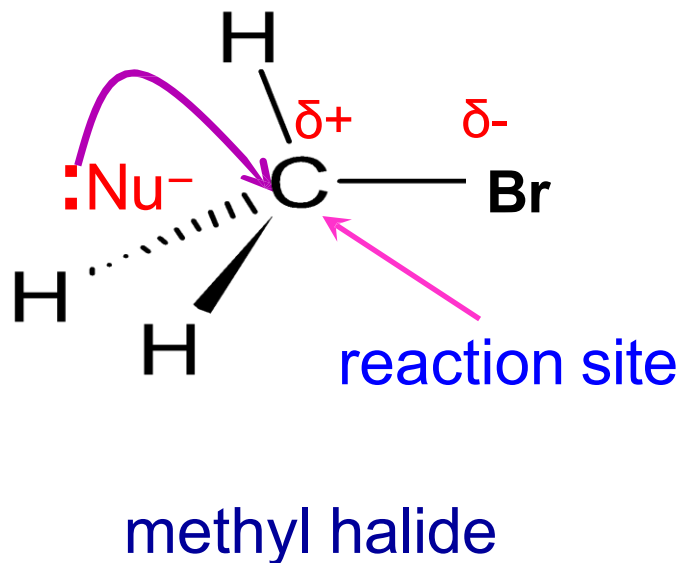


The greater the number of alkyl group or the larger the size of alkyl group attached to the α -carbon, the greater the steric effect /hindrance.

This will cause the nucleophilic attack more difficult, thus slowing the reaction rate. As a result, reactivity of haloalkane towards $\text{S}_{\text{N}}2$ decreases.



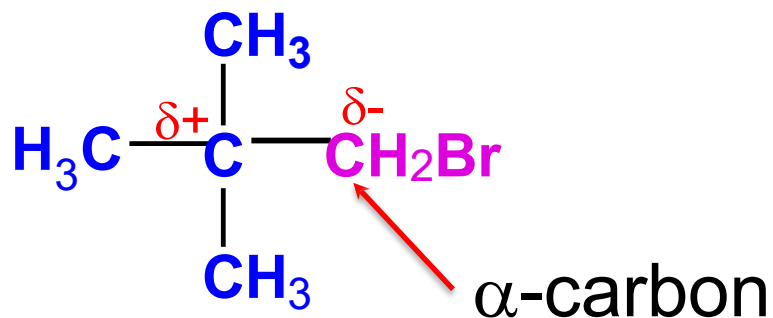
- 3° haloalkanes do NOT favour $\text{S}_\text{N}2$ reaction. 3° haloalkanes is **the least reactive** towards $\text{S}_\text{N}2$ because the **high steric effect/ hindrance** which makes nucleophilic attack difficult at α -carbon.



- In contrast, the α -carbon in methyl halide is **the least hindered @ has the lowest steric effect / hindrance**. Methyl halide is **the most reactive** towards $\text{S}_\text{N}2$.

Exceptional case

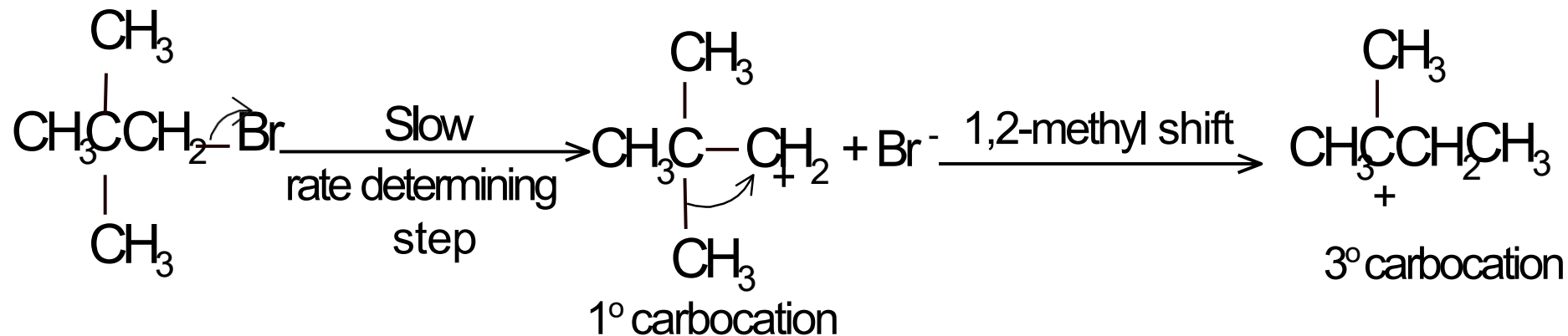
- 1° haloalkanes are not able to form stable carbocation. Therefore, generally 1° haloalkanes favour S_N2 .
- However, in some cases of 1° haloalkane such as 1-bromo-2,2-dimethylpropane (or neopentyl bromide), the presence of bulky alkyl group (tert-butyl group) makes the nucleophilic attack difficult at α -carbon.



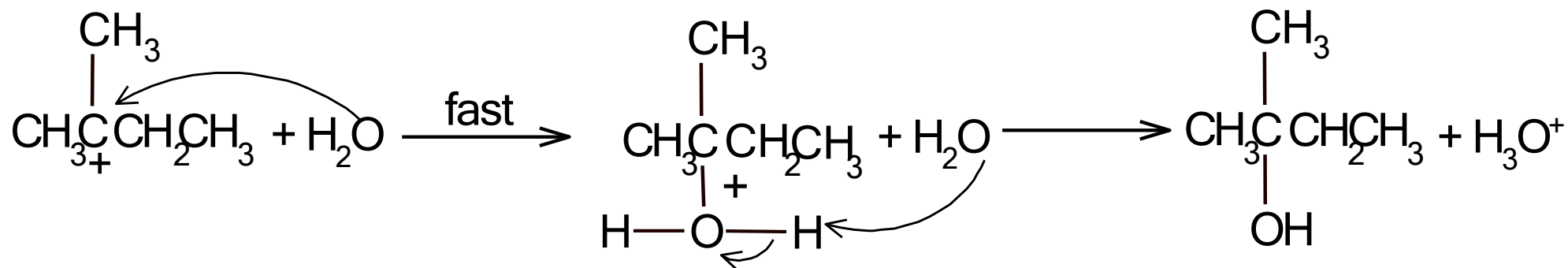
- Thus, the hydrolysis of 1-bromo-2,2-dimethylpropane occurs by S_N1 with rearrangement of carbocation. The 1° carbocation rearranges in order to form 3° carbocation which is more stable.

S_N1 mechanism with rearrangement of carbocation

Step 1: Formation of carbocation



Step 2: Nucleophilic Attack



SUMMARY OF S_N1 AND S_N2 REACTIONS

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S_N1

Unimolecular Nucleophilic Substitution

First order reaction
Rate = $k [RX]$

Order of increasing reactivity
 $CH_3X < 1^\circ < 2^\circ < 3^\circ$

Reactivity of haloalkane is
affected by carbocation stability

Multi step mechanism

S_N2

Unimolecular Nucleophilic Substitution

Second order reaction
Rate = $k [RX] [Nu:-]$

Order of increasing reactivity:
 $3^\circ < 2^\circ < 1^\circ < CH_3X$

Reactivity of haloalkane is affected
by steric effect / hindrance

One step mechanism



Analogy

S_N1 vs S_N2



STRENGTH OF NUCLEOPHILE

- Is a measure of how fast a nucleophile displaces a leaving group.
 - The S_N2 reaction depends on both the concentration of nucleophile and the identity of nucleophile.
 - A negatively charged nucleophile is generally stronger or more reactive than a similar neutral nucleophile.
- $$-OH > H_2O$$
- $$CH_3O^- > CH_3OH$$
- Strong** nucleophile in high concentration **favours** S_N2 reactions

Some Common Nucleophiles

Strong Nu:-

$-OH$, RO^- , $-CN$

Moderate:-

NH_3 , $RCOO^-$

Weak Nu:-

H_2O , ROH
 $RCOOH$

SUMMARY OF S_N1 AND S_N2 REACTIONS

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Methyl halides	1° alkyl halides	2° alkyl halides	3° alkyl halides
undergo S _N 2 only	always react by S _N 2	react by ⇒ S _N 1 in the presence of a weak nucleophile (e.g. H ₂ O, ROH) ⇒ S _N 2 in the presence of a strong nucleophile (e.g. OH ⁻ , CN ⁻ and RO ⁻)	always react by S _N 1
⇒ do not favour S _N 1 because cannot form stable carbocation. ⇒ favour S _N 2 because of the low steric effect at α-carbon.			⇒ favour S _N 1 because can form stable 3° carbocation. ⇒ do not favour S _N 2 because of the high steric effect at α-carbon.

THE USE OF HALOALKANES

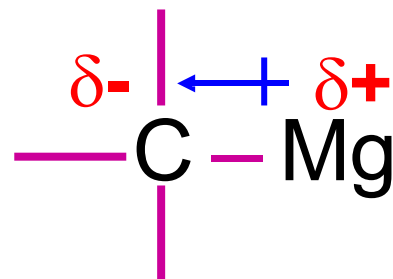
44

- Used in preparation of organometallic compound
e.g. Grignard reagents (alkyl or aryl magnesium halides)
- Grignard reagent is prepared by the reaction of an organic halide with Mg in dry ether solvent, in which the C – X bond is replaced by C – Mg bond in the reaction.

General equation:



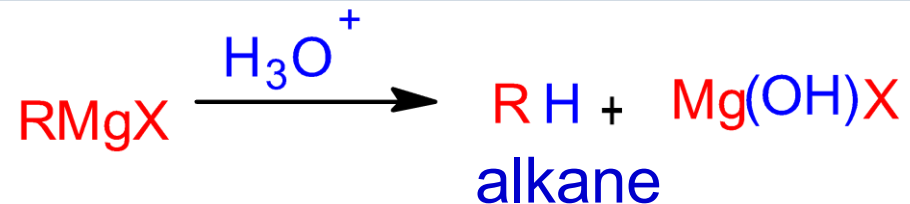
- Grignard reagent contains very polar C – Mg bond and useful in the synthesis of alkanes, alcohols and carboxylic acids.



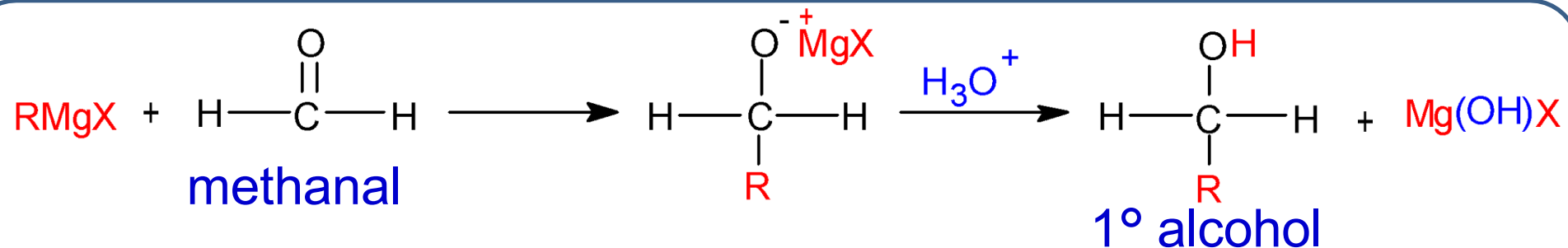
USES OF RMgX

45

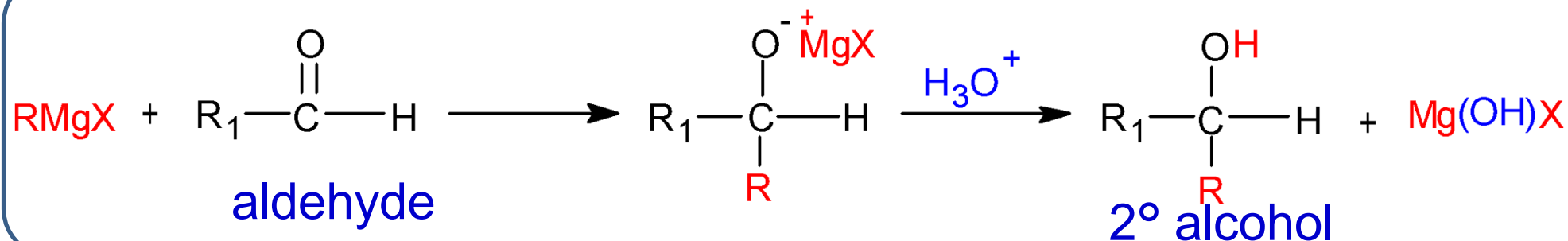
Preparation of
Alkane



Preparation of
1° Alcohol



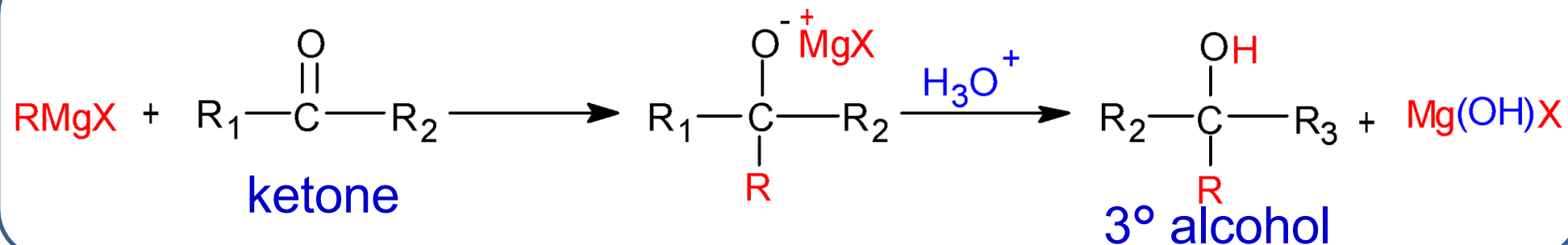
Preparation of
2° Alcohol



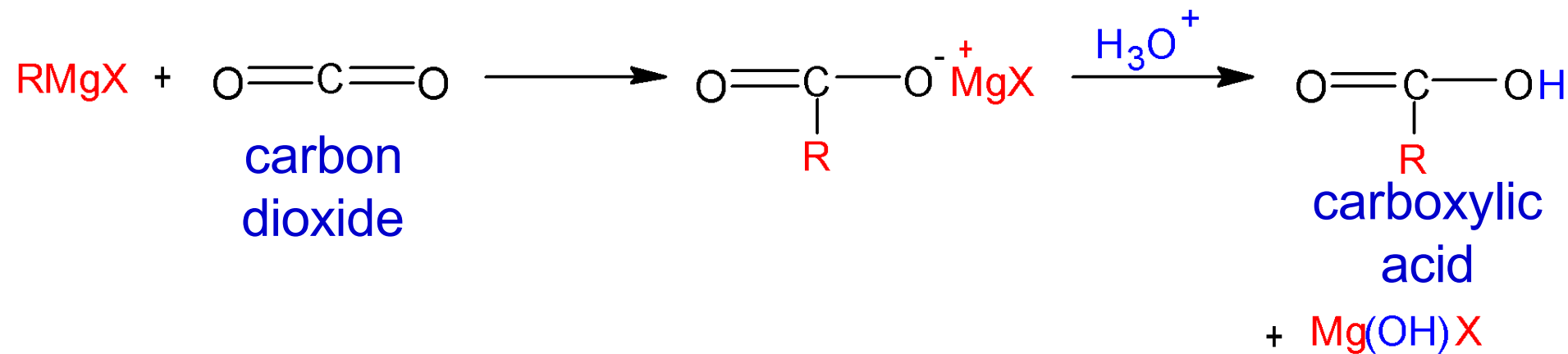
USES OF RMgX

46

Preparation of
3° Alcohol



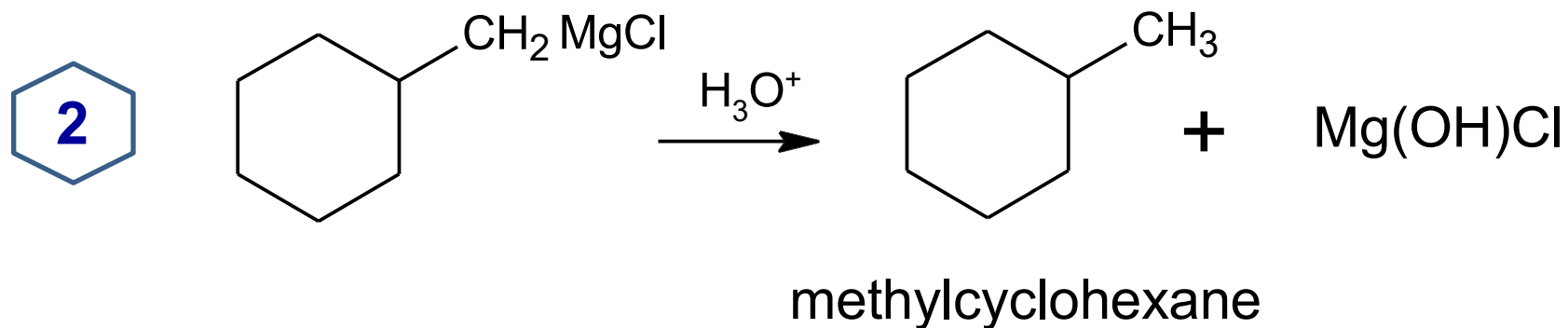
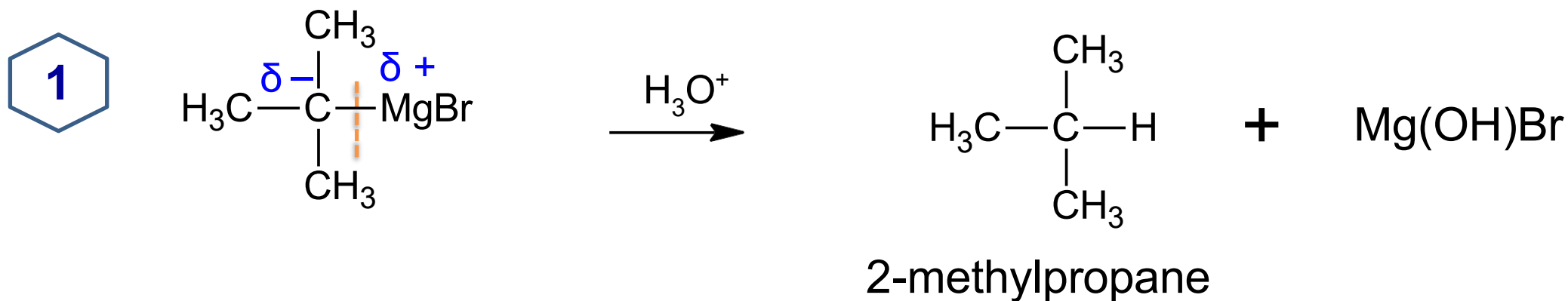
Preparation of
Carboxylic
Acid



1. PREPARATION OF ALKANES

- Acidic hydrolysis of **Grignard reagent** gives **alkane**.

Example :



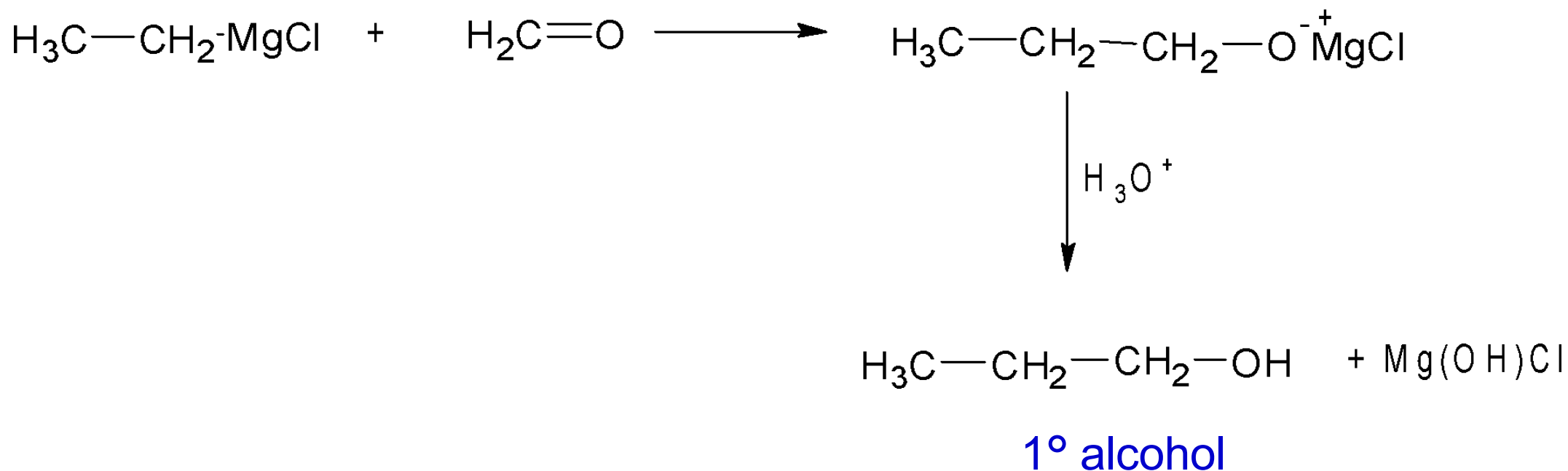
2. PREPARATION OF ALCOHOLS

48

PREPARATION OF 1° ALCOHOLS

- **Addition** of **Grignard reagent** to **methanal** followed by **hydrolysis** produces a **primary alcohol**.

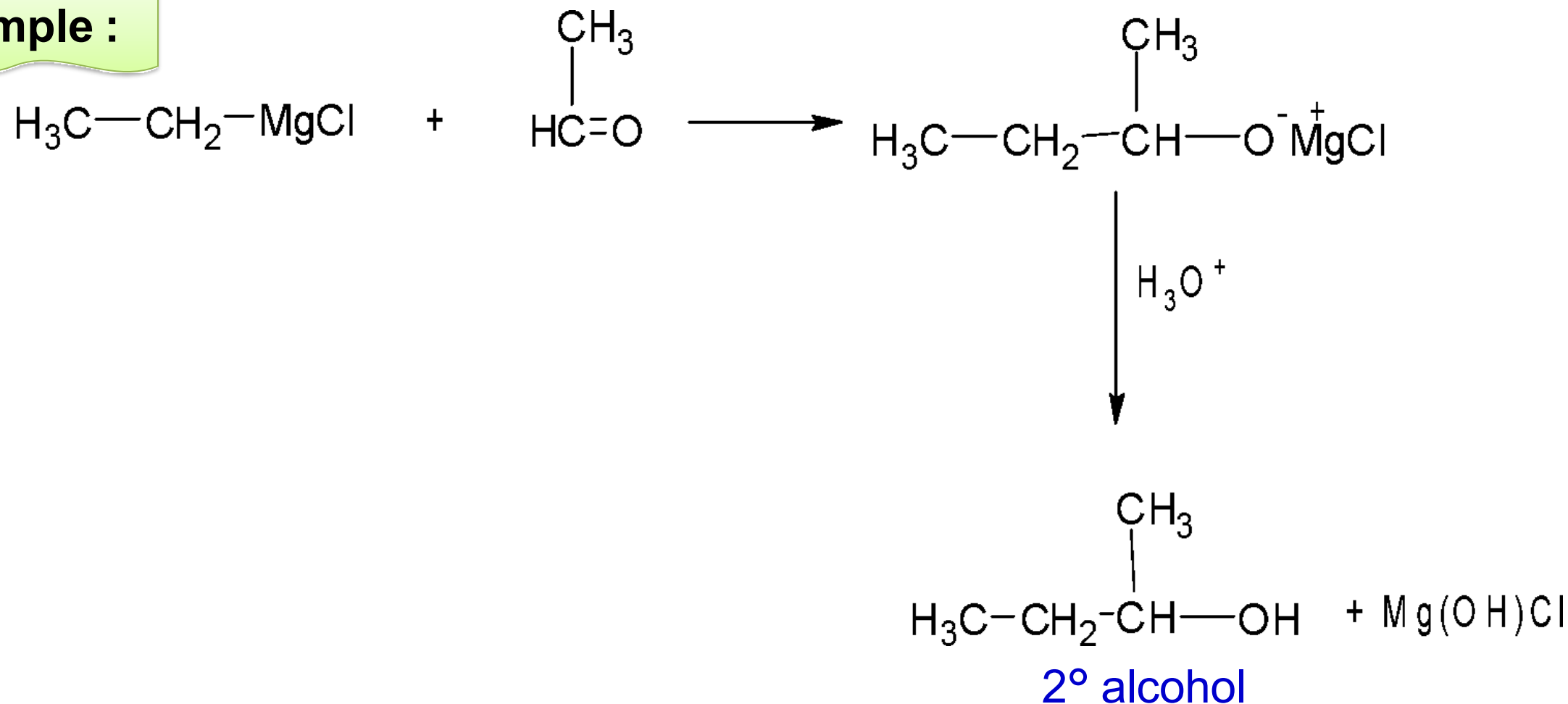
Example :



PREPARATION OF 2° ALCOHOLS

- **Addition** of **Grignard reagent** to **aldehyde** followed by **hydrolysis** produces a **secondary alcohol**.

Example :

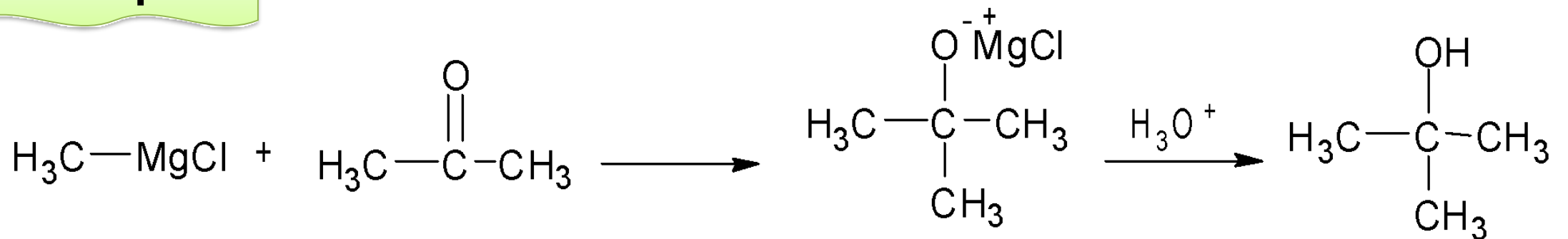


PREPARATION OF 3° ALCOHOLS

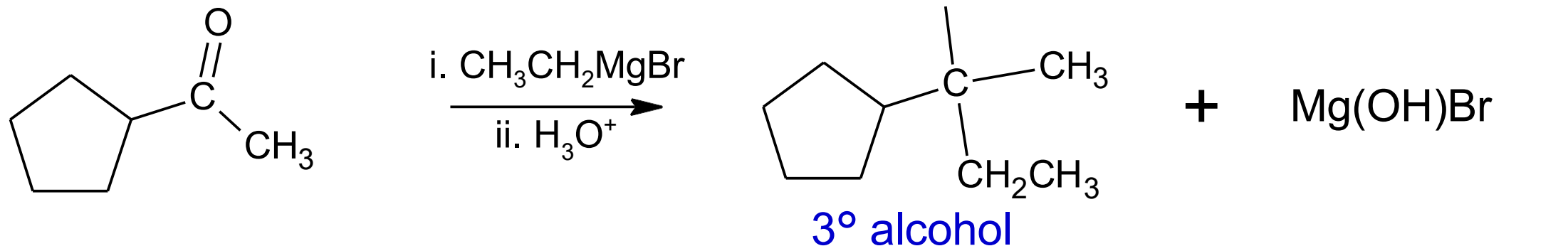
- **Addition** of **Grignard reagent** to **ketone** followed by **hydrolysis** produces a **tertiary alcohol**.

Example:

1

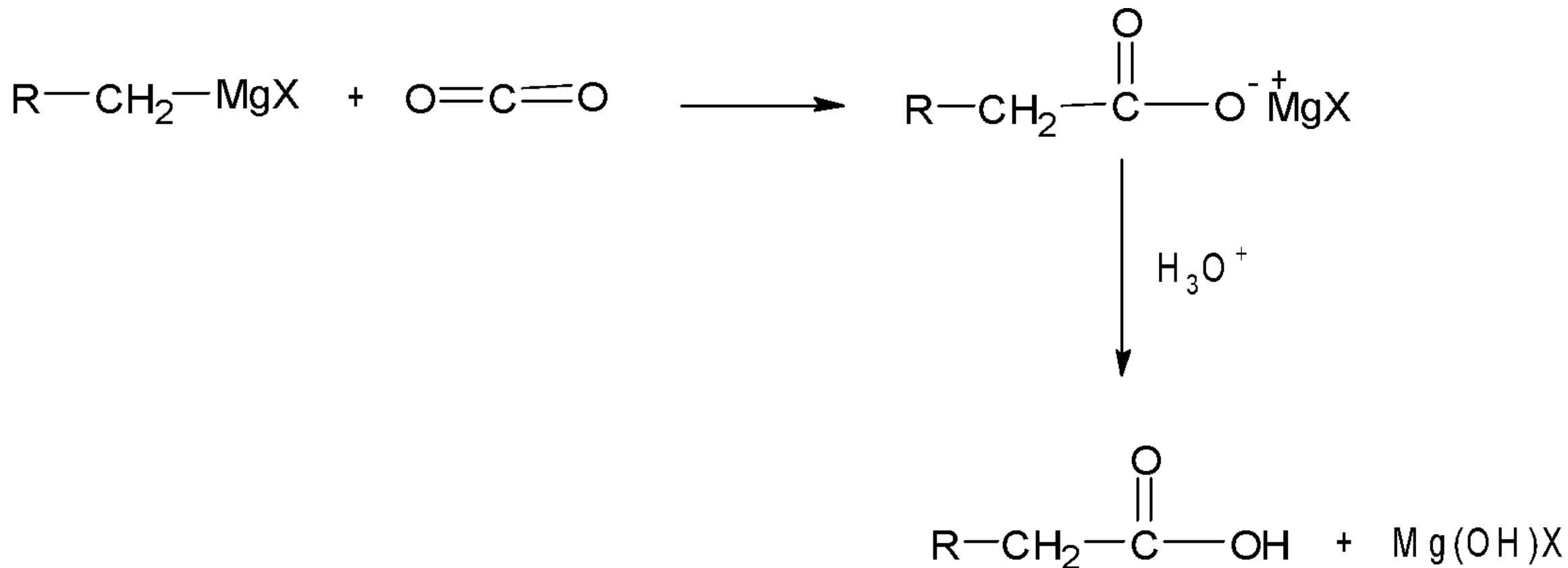


2

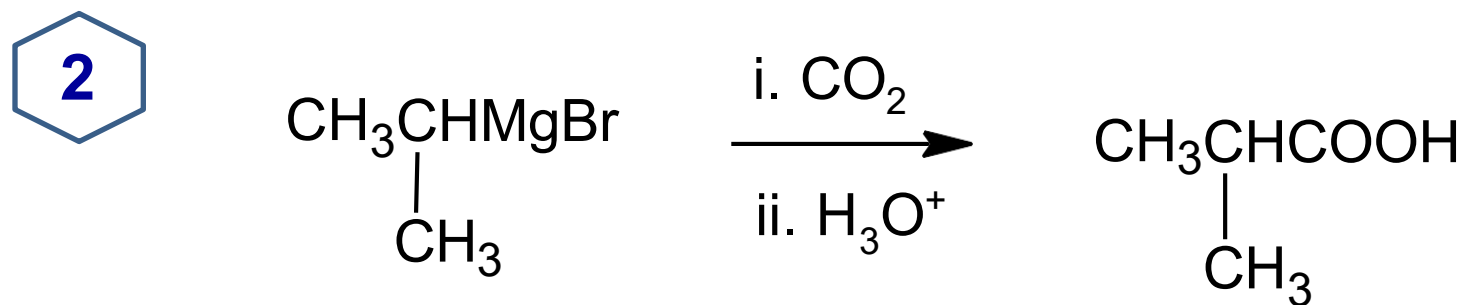
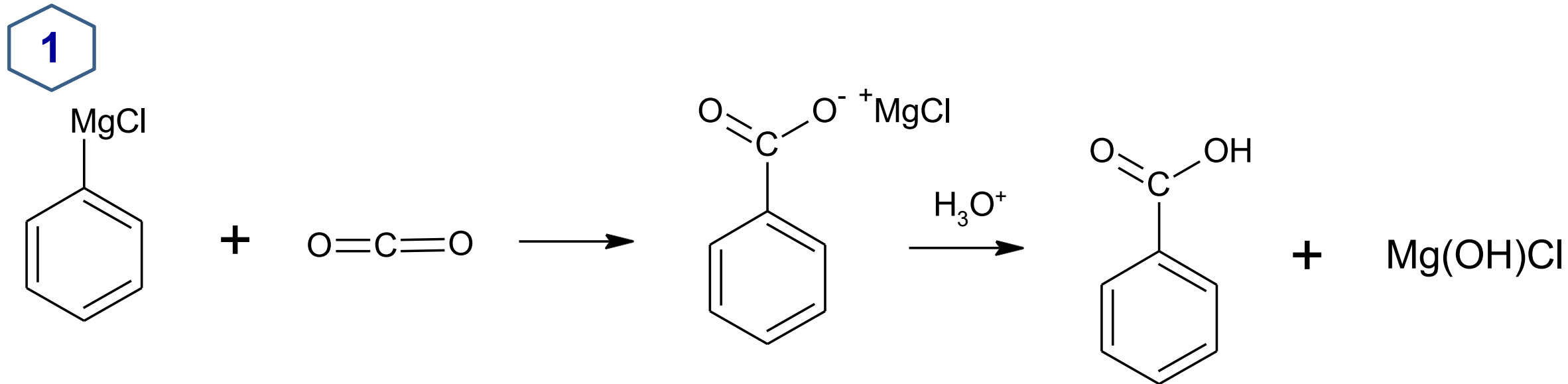


PREPARATION OF CARBOXYLIC ACID

- Reaction of **Grignard reagent** with **carbon dioxide followed by hydrolysis** produces a **carboxylic acid**.

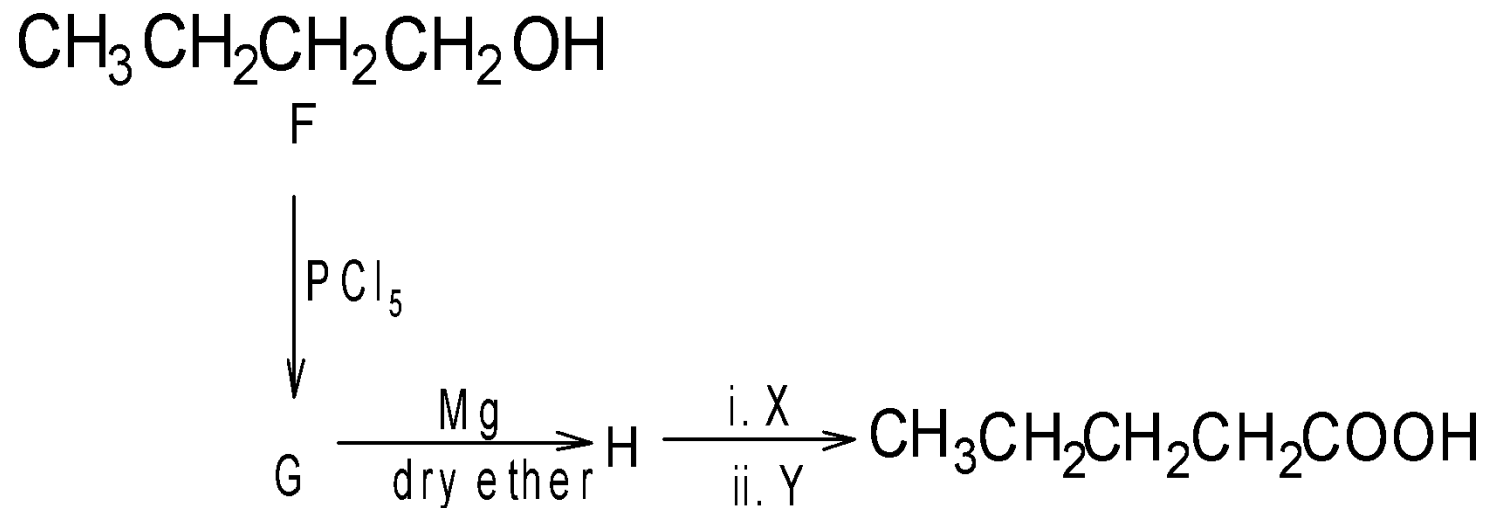


Example:



EXERCISE :

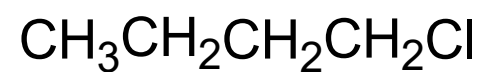
Pentanoic acid can be prepared from 1-butanol, **F** as shown below:



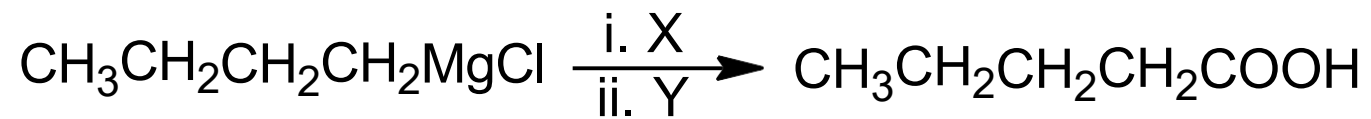
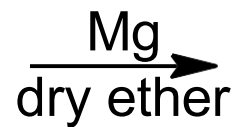
- (i) Draw the structures of **G** and **H**
- (ii) State reagents **X** and **Y**
- (iii) Give the reagent needed to prepare **F** from 1-butene.

ANSWER :

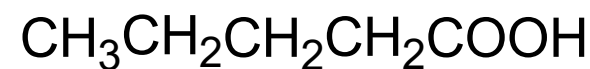
i



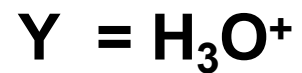
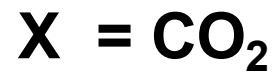
G

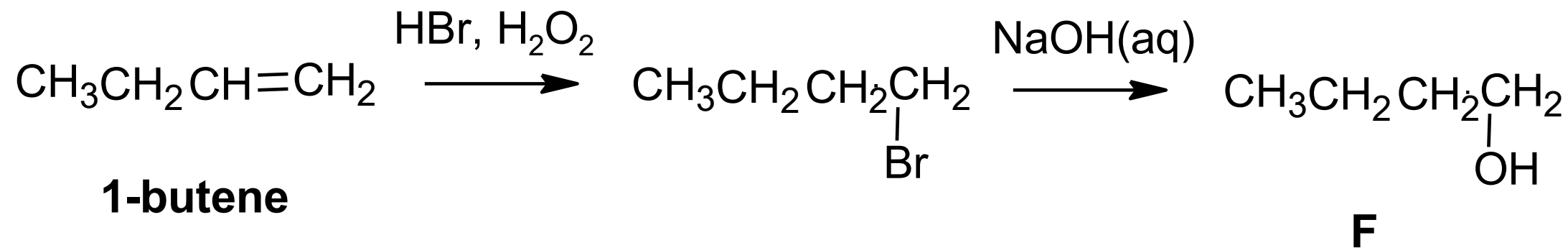
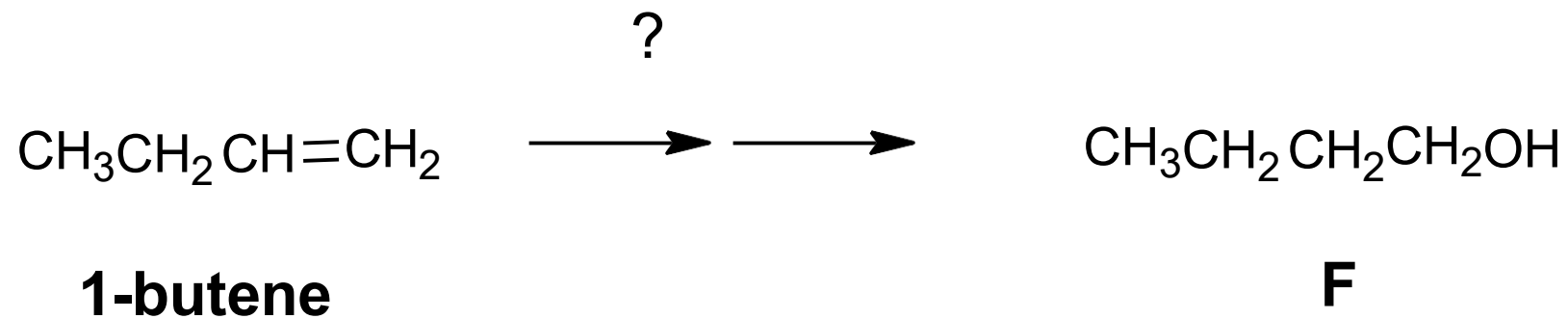


H



ii





The slide features abstract geometric designs in the corners. The top-left corner is filled with overlapping triangles in shades of blue, green, and red. The bottom-right corner contains a cluster of overlapping triangles in various shades of gray.

THE END